

Mechanics of Advanced Materials and Structures

ISSN: 1537-6494 (Print) 1537-6532 (Online) Journal homepage: www.tandfonline.com/journals/umcm20

Hybridization of generative design and TPMS lattices: A statistical approach for lightweight aeronautical components

Igor de Magalhães Catini & Guilherme Ferreira Gomes

To cite this article: Igor de Magalhães Catini & Guilherme Ferreira Gomes (2026) Hybridization of generative design and TPMS lattices: A statistical approach for lightweight aeronautical components, Mechanics of Advanced Materials and Structures, 33:1, 2692514, DOI: 10.1080/15376494.2026.2692514

To link to this article: https://doi.org/10.1080/15376494.2026.2692514

Published online: 24 Jun 2026.

Submit your article to this journal

Article views: 1

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=umcm20

MECHANICS OF ADVANCED MATERIALS AND STRUCTURES
2026, VOL. 33, NO. 1, 2692514
https://doi.org/10.1080/15376494.2026.2692514

Taylor & Francis
by informa

ORIGINAL ARTICLE

Check for updates

Hybridization of generative design and TPMS lattices: A statistical approach for lightweight aeronautical components

Igor de Magalhães Catini and Guilherme Ferreira Gomes

Mechanical Engineering Institute, Federal University of Itajubá – UNIFEI, Itajubá, Brazil

ABSTRACT

In the aerospace industry, reducing component mass without compromising mechanical performance is essential. Generative Design (GD), coupled with the geometric freedom of Additive Manufacturing (AM) and lattice structures, enables lightweight, high-strength geometries. However, lattice parameters are often selected empirically, lacking robust statistical analysis. This study addresses this gap by evaluating the mechanical response of lattice specimens across varying cell topologies, strut diameters, and materials using static Finite Element Analysis (FEA). Through Analysis of Variance (ANOVA) and multi-objective optimization, the ideal structural compromise was identified. Results demonstrated that the Gyroid structure, manufactured in AlSi10Mg with a 3 mm thickness, yields the best overall performance. To validate this methodology, a hybrid aeronautical bracket (combining a generative shell with a lattice infill) was virtually tested under operational boundary conditions. The optimized component achieved a 65.5% mass reduction compared to the original solid design for conventional manufacturing. Ultimately, the proposed hybrid approach maintains structural safety and demonstrates economic viability for AM.

ARTICLE HISTORY
Received 27 December 2025
Accepted 15 June 2026

KEYWORDS
Lattice structures; generative design; structural optimization; additive manufacturing; finite element analysis

Nomenclature

Acronyms

AM	Additive Manufacturing
ANOVA	Analysis of Variance
BCC	Body-Centered Cubic
DoE	Design of Experiments
FEA	Finite Element Analysis
FoS	Factor of Safety
GA	Genetic Algorithm
GD	Generative Design
GE	General Electric
L-PBF	Laser Powder Bed Fusion
SIMP	Solid Isotropic Material with Penalization
SLM	Selective Laser Melting
STL	Stereolithography
TO	Topology Optimization
TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution
TPMS	Triply Periodic Minimal Surfaces
UTS	Ultimate Tensile Strength

Symbols

C	Compliance
C_1, n	Gibson-Ashby constants
E	Elastic modulus (Young's modulus)
E^*	Lattice elastic modulus
E_s	Solid material elastic modulus
E_0	Solid material modulus (SIMP)
F	Applied Force
f	Target volume fraction
$\mathbf{K}$	Global stiffness matrix
$\mathbf{k}_0$	Element stiffness matrix
L	Unit cell size
M	Moment
p	Penalization factor
R^2	Coefficient of determination
t	Isosurface level / Beam thickness
$\mathbf{U}$	Global displacement vector
$\mathbf{u}_e$	Element displacement vector
V	Volume
x_e	Element relative density
$x_{\min}$	Minimum relative density vector
ν	Poisson's ratio
ρ	Specific mass / Density
$\overline{\rho}$	Relative density
ρ^*	Cellular structure density
ρ_s	Solid material density
σ^*	Yield strength

1. Introduction

The current landscape of mechanical and aerospace engineering is marked by the relentless pursuit of energy efficiency and structural performance. In this context, mass reduction (lightweighting) has become a mandatory requirement, driving the development of new optimization techniques that transcend the limitations of conventional manufacturing [1,2]. Topology Optimization (TO) and, more recently, Generative Design (GD), have established themselves as essential tools for distributing material only where it is strictly necessary [3,4]. However, the geometric complexity resulting from these approaches often precludes

CONTACT Guilherme Ferreira Gomes guilhermefergom@gmail.com Mechanical Engineering Institute, Federal University of Itajubá – UNIFEI, Itajubá, Brazil.
© 2026 Taylor & Francis Group, LLC

production by subtractive methods, finding in Additive Manufacturing (AM) the ideal means for their realization [5,6].

Additive Manufacturing, by allowing layer-by-layer fabrication, has opened doors for the utilization of cellular and reticulated structures, known as Lattices. According to [7], these structures, inspired by natural materials such as bone and wood, offer a high strength-to-weight ratio and excellent energy absorption capabilities. Recent studies have demonstrated that replacing solid volumes with lattice structures can drastically reduce component mass without compromising its integrity, provided that geometric parameters are correctly tuned [8,9].

The materials selected for this study were aluminum AlSi10Mg, titanium Ti6Al4V, and 316 L stainless steel. The choice of these materials allows for the evaluation of the effectiveness of lattice structures across a wide range of densities and specific strengths: AlSi10Mg offering an excellent specific strength ratio; Ti6Al4V providing superior performance in specific strength; while 316 L steel provides a benchmark for high ductility and lower cost [6,10].

In the realm of cellular topologies, Triply Periodic Minimal Surfaces (TPMS), such as the gyroid structure, have gained prominence in the literature. Unlike beam-based lattices (strut-based), TPMS feature zero mean curvature and a lack of sharp nodes, which reduces stress concentration and improves fatigue resistance and manufacturability [11,12]. Research conducted by [13] and [14] has validated the use of alloys such as AlSi10Mg and Ti6Al4V for these structures, demonstrating that the microstructure of the material processed by Selective Laser Melting (SLM) plays a crucial role in final performance.

However, the literature points out challenges in the integration of these technologies. While [15] propose hybrid shell and infill methods (shell-infill) for optimization [16], discuss the computational limitations of Generative Design in automatically creating complex internal structures [17]. warn that, although software can generate organic shapes, the definition of unit cell parameters (type, diameter, relative density) is often done empirically or based on trial and error, lacking a robust statistical analysis that correlates process variables with mechanical behavior. Furthermore [18], highlight the need to evaluate not only mechanical performance but also the economic and manufacturing viability of these complex geometries.

Despite the significant advances reported by [19] and [20] in lattice simulation, there is a gap in the literature regarding the systematic application of statistical methods (such as DOE and ANOVA) to optimize TPMS lattices specifically integrated into geometries resulting from Generative Design for complex loading cases, such as aeronautical brackets.

In light of the above, the present work seeks to address this gap in the scientific literature. Unlike approaches that apply generic lattices to primitive geometries, this study proposes an integrated methodology that unites Generative Design with the statistical optimization of cellular structures. The main contribution lies in the determination, *via* sensitivity analysis and multi-objective optimization, of the ideal topology and diameter configuration for filling a real aeronautical component, proving that the hybridization of a generative shell with an optimized gyroid core surpasses conventional solutions in terms of structural efficiency and mass reduction.

Table 1. Properties of the materials used.

	Material		
Property	316L	AlSi10Mg	Ti6Al4V
Poisson's Ratio (ν)	0.30	0.33	0.32
Elastic Modulus (E) [GPa]	200	70	117
Ultimate Tensile Strength [MPa]	595	470	1040
Density (ρ) [kg/m^3]	8000	2660	4430

2. Theoretical background

This Section presents the fundamental concepts underpinning the development of this work, addressing Additive Manufacturing, the mechanics of Cellular Structures (Lattices), and the mathematical principles of Generative Design and Topology Optimization.

2.1. Additive manufacturing

Additive Manufacturing (AM), popularly known as 3D printing, is defined by Gibson et al. [5] as the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing methodologies. According to Volpato [6] and others ([21]), this technology allows for the fabrication of complex geometries that would be infeasible or excessively costly *via* traditional methods, such as curved internal channels and reticulated structures.

The geometric freedom provided by AM fostered the concept of "complexity for free", where the manufacturing cost is relatively independent of the part's shape complexity, depending mainly on the material volume and machine time [17].

The material selection in Additive Manufacturing was conducted as follows: AlSi10Mg is an aluminum alloy that, in AM, presents a refined microstructure and high mechanical strength combined with low density, making it ideal for weight reduction components [22]. The titanium alloy Ti6Al4V stands out for its corrosion resistance and high strength-to-weight ratio, being the standard material for aerospace structural components despite its high cost [23]. Finally, 316 L stainless steel is widely employed due to its high ductility, toughness, and oxidation resistance, serving as a robust and comparatively lower-cost reference material for validating complex geometries [24]. Table 1 shows the used properties applied in this work.

The cost of materials is justified by the Laser Powder Bed Fusion (L-PBF) manufacturing method. In this process, the price is hardly affected by the time required for the machine to produce the part. Therefore, the cost model was based on the price of the material [25] and the cost of production [26], which includes energy, production time, setup, and post-processing. This amount provides the estimated final cost per kilogram (Table 2).

Table 2. Estimated cost breakdown per kilogram for L-PBF manufacturing.

Material	Powder Cost [$/kg]	Production Cost [$/kg]	Final Cost [$/kg]
316 L	85.00	465.00	550.00
AlSi10Mg	90.00	680.00	770.00
Ti6Al4V	350.00	1750.00	2100.00

2.2. Lattice structures

Cellular solids, or lattice structures, are configured by the repetition of a unit cell in space, forming an interconnected network of beams or surfaces [27]. According to Gibson and Ashby [7], the most important physical property of these structures is their Relative Density ($\overline{\rho}$), defined as the ratio between the density of the cellular structure (ρ^*) and the density of the solid material comprising it (ρ_s), as per Equation 1:

$$\overline{\rho} = \frac{\rho^*}{\rho_s} \tag{1}$$

The mechanical properties of the lattice, such as the Elastic Modulus (E^*) and Yield Strength (σ^*), degrade as relative density decreases. Gibson and Ashby [7] propose a power-law relationship to describe this behavior, determined by Equation 2:

$$\frac{E^*}{E_s} = C_1 \left(\frac{\rho^*}{\rho_s}\right)^n \tag{2}$$

where E_s is the solid material modulus, and C_1 and n are constants dependent on the unit cell topology.lattice structures can be classified, according to their connectivity and geometry, into two major groups: beam-based (strut-based) and surface-based (TPMS).

TPMS are composed of nodes connected by cylindrical or prismatic rods. Common examples include the Body-Centered Cubic (BCC) cell, the Octet-truss, and the Kelvin cell. According to Maskery et al. [9], the mechanical behavior of these structures (stretching- or bending-dominated) is determined by the connectivity number at the nodes (Maxwell's Criterion). Although efficient, they present geometric discontinuities at the nodes, which act as stress concentrators under loading and can accelerate fatigue failure, according to Panesar et al. [8].

TPMS are smooth, continuous, non-self-intersecting surfaces that divide space into two intertwined labyrinths. The gyroid structure, the focus of this work, is a prominent example of TPMS. Al-Ketan and Abu Al-Rub [11] highlight that, unlike strut-based trusses, TPMS possess zero mean curvature at all points.

The approximate equation for the gyroid surface is given by Equation 3:

$$\sin\left(\frac{2\pi x}{L}\right)\cos\left(\frac{2\pi y}{L}\right) + \sin\left(\frac{2\pi y}{L}\right)\cos\left(\frac{2\pi z}{L}\right) + \sin\left(\frac{2\pi z}{L}\right)\cos\left(\frac{2\pi x}{L}\right) = t \tag{3}$$

where L is the unit cell size and t is the isosurface level controlling relative density. According to Yan, Rao, and Li [20], the absence of sharp nodes in TPMS results in a more uniform stress distribution and better manufacturability in 3D printing, as they often dispense with internal supports (self-supporting).

This self-supporting characteristic is a decisive factor for the reliability of the mechanical results presented. The Gyroid geometry employed in this study relies on continuous curvature where the local overhang angles remain above the critical threshold for Laser Powder Bed Fusion (typically 45°), effectively eliminating the need for internal auxiliary structures. This is particularly important for lattice integrity, as the generation of supports within the intricate, sub-millimeter voids of the cells would render their removal physically impossible during post-processing. Consequently, any reliance on internal supports would result in "trapped" material, which would artificially increase the relative density and stiffness of the specimens, introducing a significant discrepancy between the simulated nominal geometry and the manufactured component.

2.3. Generative design and topology optimization

Generative Design is an iterative form-finding process that utilizes artificial intelligence and cloud computing to generate multiple design solutions meeting specific constraints, as defined by Buonamici et al. [4]. The fundamental mathematical basis of Generative Design is Topology Optimization (TO).

Topology Optimization seeks the optimal material distribution within a fixed design domain to minimize an objective function (usually flexibility or compliance) subject to volume constraints. The most widespread method in the literature is SIMP (Solid Isotropic Material with Penalization), described by Bendsøe and Sigmund [3].

In the SIMP method, the relative density (x_e) of each finite element can vary continuously between 0 (void) and 1 (solid). The relationship between element density and its elastic modulus is given by Equation 4:

$$E(x_e) = (x_e)^p E_0 \tag{4}$$

where E_0 is the solid material modulus and p is the penalization factor (usually $p = 3$) to force the solution to converge to discrete values of 0 or 1.

Mathematically, the optimization problem to minimize compliance (C) is formulated as Equation 5:

$$\begin{aligned} &\text{Minimize}_{\mathbf{x}}: && C(\mathbf{x}) = \mathbf{U}^T\mathbf{K}\mathbf{U} = \sum_{e=1}^{N}(x_e)^p \mathbf{u}_e^T \mathbf{k}_0 \mathbf{u}_e \\ &\text{Subject to}: && \frac{V(\mathbf{x})}{V_0} = f \\ & && 0 < x_{min} \le x_e \le 1 \end{aligned} \tag{5}$$

where $\mathbf{U}$ is the global displacement vector, $\mathbf{K}$ is the global stiffness matrix, $\mathbf{u}_e$ and $\mathbf{k}_0$ are the element displacement and stiffness, $V(\mathbf{x})$ is the material volume, and f is the target volume fraction, according to Bendsøe and Sigmund [3].

In the present work, Generative Design is used to create the macrostructure (shell and main load paths), while lattice structures fill the internal volume, creating a hybrid material with optimized properties.

3. Methodology

3.1. Modeling and simulation of lattice structures

In order to investigate the mechanical behavior and properties of different lattice structure topologies, static simulations were conducted using the Finite Element Method (FEA). The central objective consisted of evaluating and selecting the structure with superior performance by varying only the geometric parameters of the unit cell type and the beam diameter or wall thickness. Based on the results obtained, Design of Experiments (DoE) techniques were applied allied with mono- and multi-objective optimizations to identify optimal solutions. The selected configuration was then integrated into a component developed *via* Generative Design, aiming to validate the proposed methodology. This validation was based on a comparative analysis of mass reduction and structural stiffness in relation to the original component, which was designed for conventional manufacturing.

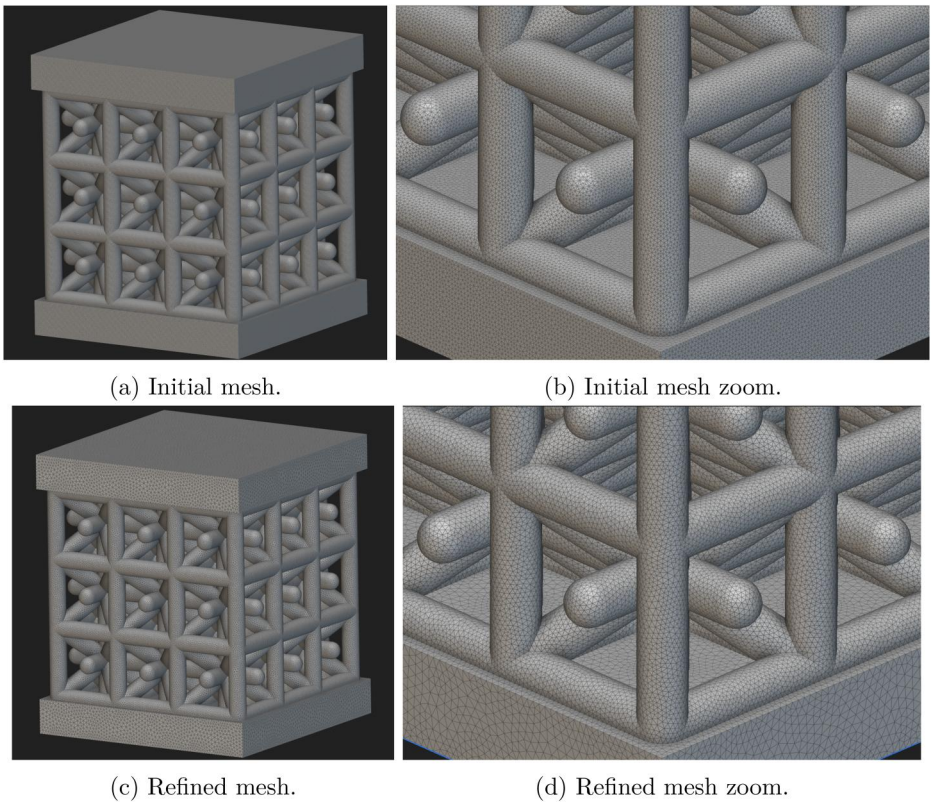
The computational modeling of the specimens was developed within the nTop software environment, using field-based modeling resources (Implicit Bodies) and lattice structures. The volume of the cellular structure was delimited by two solid plates, upper and lower, responsible for the application of the mechanical load and the restriction of degrees of freedom (displacement), respectively.

Aiming at process optimization, an automated workflow was structured to allow for agile parametric variation of the geometry. The cell topology and beam diameter were defined as input variables, while material properties were configured as workflow constants. This systematic approach makes the analysis of different thicknesses and cell types efficient, as altering a parameter automatically triggers the simulation update. The methodological procedure is therefore divided into parametric modeling of the specimen and configuration of the static finite element analysis (FEA).

The modeling stage began with the definition of the upper and lower support plates, responsible for applying the boundary conditions (loading and fixation). These plates were generated with dimensions of $68 \times 68 \times 10$ mm and positioned with a vertical spacing of 60.8 mm. The volume occupied by the lattice structure was delimited by a central block of $60 \times 60 \times 60$ mm. The discretization of this domain was configured with 3 unit cells in each direction, possessing individual dimensions of $20 \times 20 \times 20$ mm, which resulted in a matrix of $3 \times 3 \times 3$ cells, totaling 27 units.

The strut or wall thickness of the unit cells was parameterized, enabling the agile adjustment of the structure's relative density for different test scenarios. The final geometric model was consolidated through a Boolean union operation between the generated lattice volume and the support plates. This procedure ensured the elimination of redundant intersections between bodies, preventing inconsistencies during the subsequent mesh generation, as illustrated in Figure 1. The types of unit cells tested are represented in Figure 2 and the arrangements in Figure 3.

The structural analysis integrated the discretized model with its physical properties and boundary conditions. To execute the simulation, a surface mesh was initially generated from the implicit geometry, followed by the creation of the volumetric mesh to accurately define the regions for constraints and loads.



(a) Initial mesh. (b) Initial mesh zoom.

(c) Refined mesh. (d) Refined mesh zoom.

Figure 1. Lattice specimen meshes.

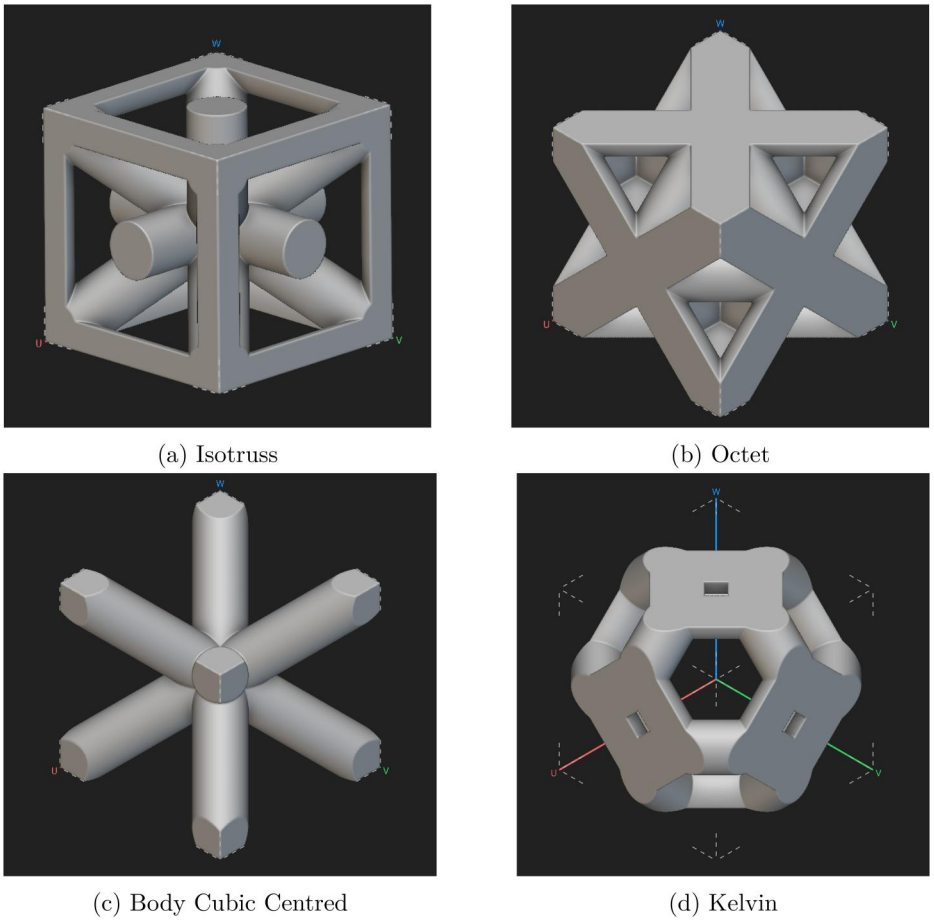


Figure 2. Cell types.

Subsequently, the solid model was consolidated by associating the volumetric mesh with the specified material.

The mechanical properties assigned to the materials were selected based on literature standards for components processed *via* Laser Powder Bed Fusion (L-PBF). However, for the purpose of these numerical simulations, the materials were modeled as continuous, homogeneous, and isotropic. While it is well-documented that the layer-by-layer nature of additive manufacturing induces microstructural anisotropy—resulting in mechanical variations depending on the build orientation (Z-direction vs. X/Y-plane), this study assumes isotropy as a valid approximation for the comparative analysis of lattice topologies. This simplification is necessary to maintain computational viability during the iterative optimization process, with the understanding that the values used represent an average of the nominal properties expected for as-printed parts.

The initial surface discretization was configured with a tolerance of 0.8 mm (representing the maximum permitted deviation of the mesh relative to the ideal implicit geometry) and a minimum element size of 0.4 mm to preserve smaller geometric details and avoid self-intersections. This parameterization produced an initial surface mesh ranging from 1.2 to 1.5 million elements.

To optimize computational resources without compromising accuracy, this initial mesh was refined and simplified. The growth rate was set to 1.2, allowing adjacent elements to be up to 20% larger in regions of lower curvature. The chord height was defined as 0.02 times the mesh tolerance, and the minimum element size was restricted to 0.33 times the tolerance. This strategy significantly reduced model complexity, resulting in a refined surface mesh of between 500,000 and 800,000 elements.

To ensure the numerical accuracy of the Finite Element Analysis (FEA) results, a mesh sensitivity study was conducted specifically for the TPMS architectures. The element size was systematically reduced from 1.8 mm to 0.7 mm to monitor the convergence of the maximum Von Mises stress. As shown in Figure 4, the stress values exhibit a clear stabilization trend as the mesh is refined. For the 3 mm strut thickness, an element size of 0.4 mm was selected, providing approximately 7 to 8 elements through the thickness. This discretization level resulted in a relative variation of less than 0.5% compared to the finest mesh tested (0.7 mm), confirming that the chosen mesh density is sufficient to capture local stress concentrations and provide grid-independent results for the optimization study.

For the volumetric domain, the refined surface was used as the input to generate the volume mesh (Figure 5). The internal element size was set to 2.0 mm, with a volumetric growth rate of 1.5. Table 3 presents a summary of the mesh configurations used.

The numerical discretization was performed using 10-node quadratic tetrahedral elements to ensure a high-fidelity

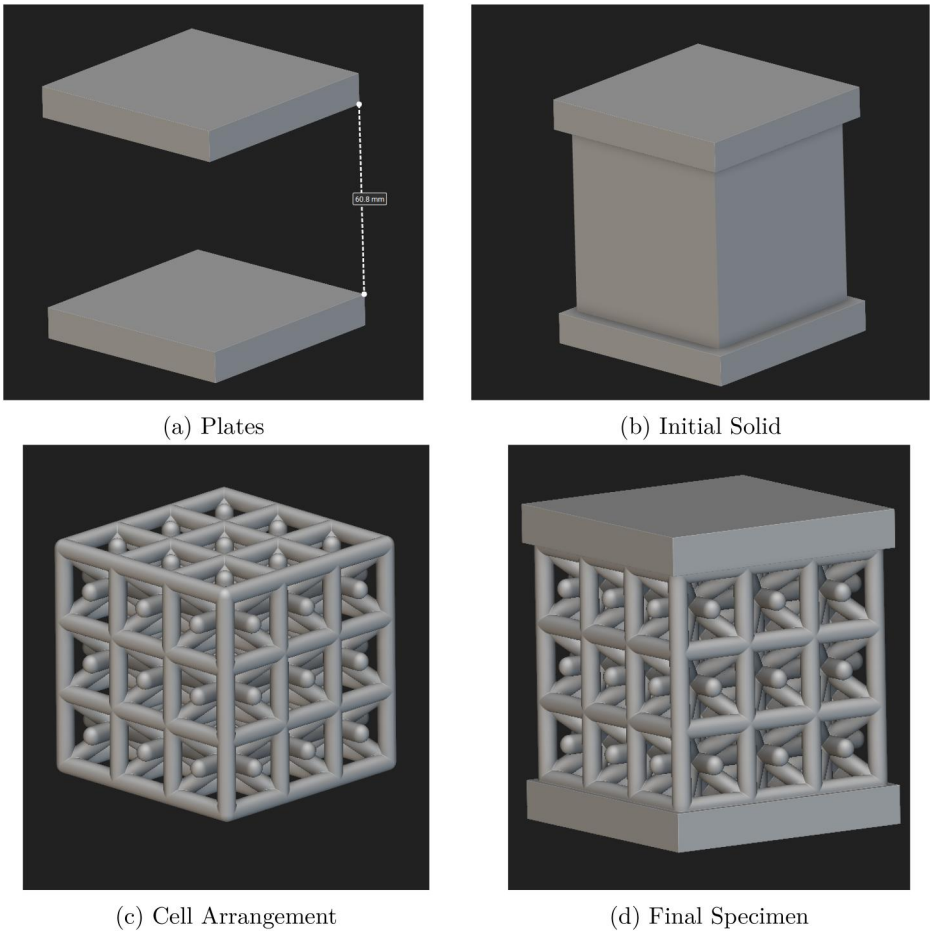


Figure 3. Lattice specimen modeling.

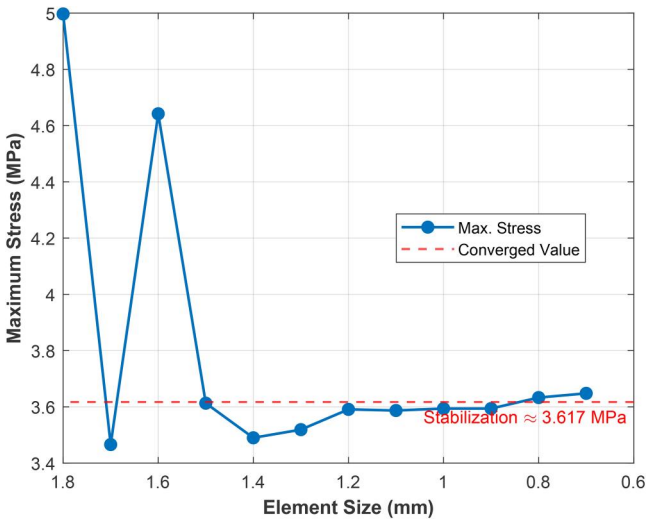


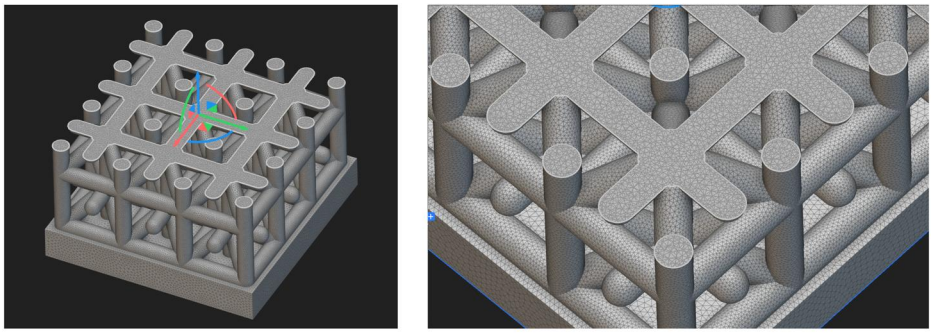
Figure 4. Mesh sensitivity analysis showing the convergence of maximum stress as a function of element size.

representation of the complex lattice curvatures. Although 1D beam or 2D shell elements could reduce computational effort, 3D solid elements were preferred to accurately capture the volumetric stress distribution at the core-shell interfaces and the local gradients within the TPMS struts. The final mesh density, determined by the convergence study, resulted in models with approximately 5,000,000 to 8,000,000 and 500,000 to 800,000 elements, ensuring that the optimization was driven by grid-independent structural responses.

Finally, the boundary conditions were established as illustrated in Figure 6. A constant compressive load of $F = 1$ kN was applied to the upper plate. On the lower plate, a perfect fixed constraint was imposed, blocking all degrees of translational and rotational freedom ($U_x = U_y = U_z = 0$ and $R_x = R_y = R_z = 0$). Simulations were then executed for each combination of material, thickness, and lattice typology to extract maximum stress, displacement, and mass data. Table 4 summarizes the applied conditions.

3.2. Generative design

The part selected for the application of the lattice structure refers to the case study proposed by General Electric (GE), named the *GE Jet Engine Bracket Challenge* [28]. In this challenge, the optimization of an existing jet engine bracket was sought, based on the use of additive manufacturing. The central objective consisted of applying optimization technologies to promote a significant mass reduction in parts originally designed for conventional manufacturing processes. In addition to the design requirements, the original geometry of the part was provided as a reference. The bracket

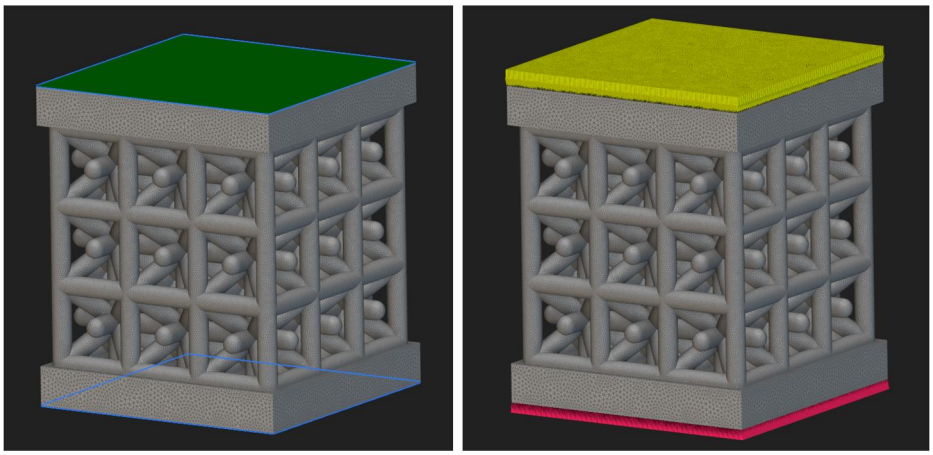


(a) Volume mesh. (b) Volume mesh zoom.

Figure 5. Volume mesh details.

Table 3. Mesh configuration parameters (initial, refined, and volume).

Stage	Parameter	Configured Value
Initial Surface Mesh	Tolerance	0.8 mm
	Min. Element Size	0.4 mm
	Sharpen Iterations	10
	Resulting Elements	1.2 to 1.5 million
Refined Surface Mesh	Growth Rate	1.2 (20%)
	Chord Height	0.02 × Tol.
	Min. Element Size	0.33 × Tol.
	Resulting Elements	500,000 to 800,000
Volumetric Mesh	Element Size	2.0 mm
	Growth Rate	1.5
	Element Order	Quadratic



(a) Constraints (b) Loading

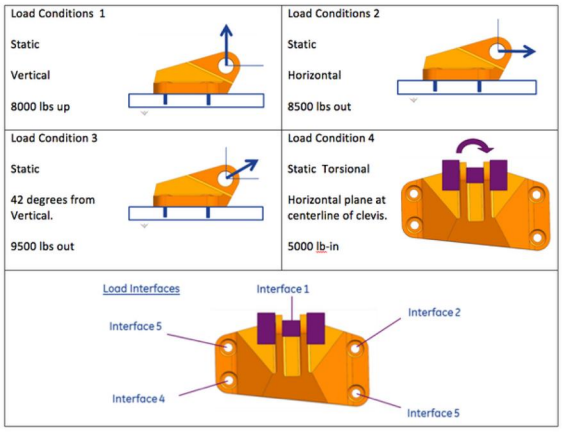
Figure 6. Boundary conditions for lattice specimen.

must support the weight of the engine and, to do so, the design must obey the boundary conditions of Figure 7 and Table 5.

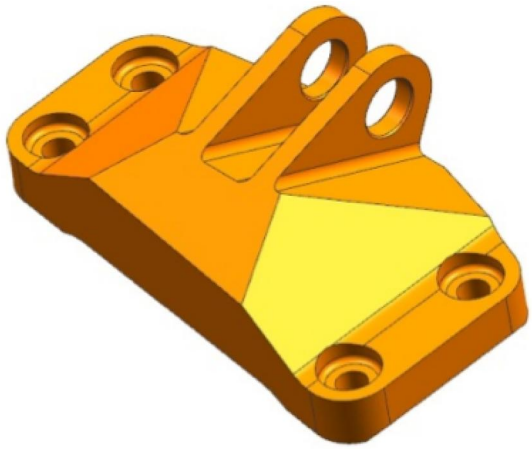
To solve the problem, the Generative Design technique was adopted using PTC Creo Parametric software. The study configuration process was based on the distinction between preservation and restriction geometries. The former represent the mandatory mechanical interfaces, such as holes and support bases, while the latter define prohibited volumes, ensuring that the resulting part does not interfere with the engine envelope.

Unlike traditional topology optimization, which requires an initial design volume to undergo mass reduction, a pure generative approach was chosen, where the algorithm explores the free space to create optimized load paths without the bias of the original geometry. Although the reference model for the boundary conditions was extracted from the Autodesk Fusion 360 environment, the analysis and generation of the final geometry were conducted in PTC Creo. This strategic decision is due to Creo's more robust architecture for high-complexity engineering, offering greater control over the quality of the generated mesh and a more precise integration with finite element analysis (FEA) tools, which are crucial factors for ensuring the reliability of critical aerospace components.

The preservation geometries were defined covering the fixed parts, corresponding to interfaces 2, 3, 4, and 5 (intended for bolting), as well as interface 1, the contact



(a) GE Bracket boundary conditions



(b) Initial GE Bracket part

Figure 7. GE bracket study (adapted from [28]).

Table 4. Boundary conditions for static simulation of specimens.

Type	Location	Description	Degrees of Freedom
Loading	Upper Plate	Constant Compressive Force	$F_y = -1000$ N (1 kN)
Constraint	Lower Plate	Fixed Support	$U_x = U_y = U_z = 0$
			$R_x = R_y = R_z = 0$

Table 5. Boundary conditions and total loading at interface 1.

		Total Load at Interface 1 (N)			Moment
Case	Description	$\mathbf{F_x}$ (Horiz.)	$\mathbf{F_y}$ (Lat.)	$\mathbf{F_z}$ (Vert.)	$\mathbf{M_z}$ (N · m)
1	Vertical Load (Up)	0	0	+35,586	–
2	Horizontal Load (Out)	+37,810	0	0	–
3	Inclined Load (42°)*	+28,276	0	+31,404	–
4	Torsional Load**	–	–	–	564.9
-	**Fixation (Interfaces 2, 3, 4, 5)**	$U_x = 0$	$U_y = 0$	$U_z = 0$	$Rot = 0$

Notes.
1. Values represent the total load applied to the pin (Interface 1).
2. (*) Decomposition of the total load of 9500 lbs: $F_z = F_{total} \cdot \cos(42^\circ)$; $F_x = F_{total} \cdot \sin(42^\circ)$.
3. (**) Torque is applied as a Moment on the central axis.
4. In constraints: $U = 0$ indicates null displacement (fixed support).

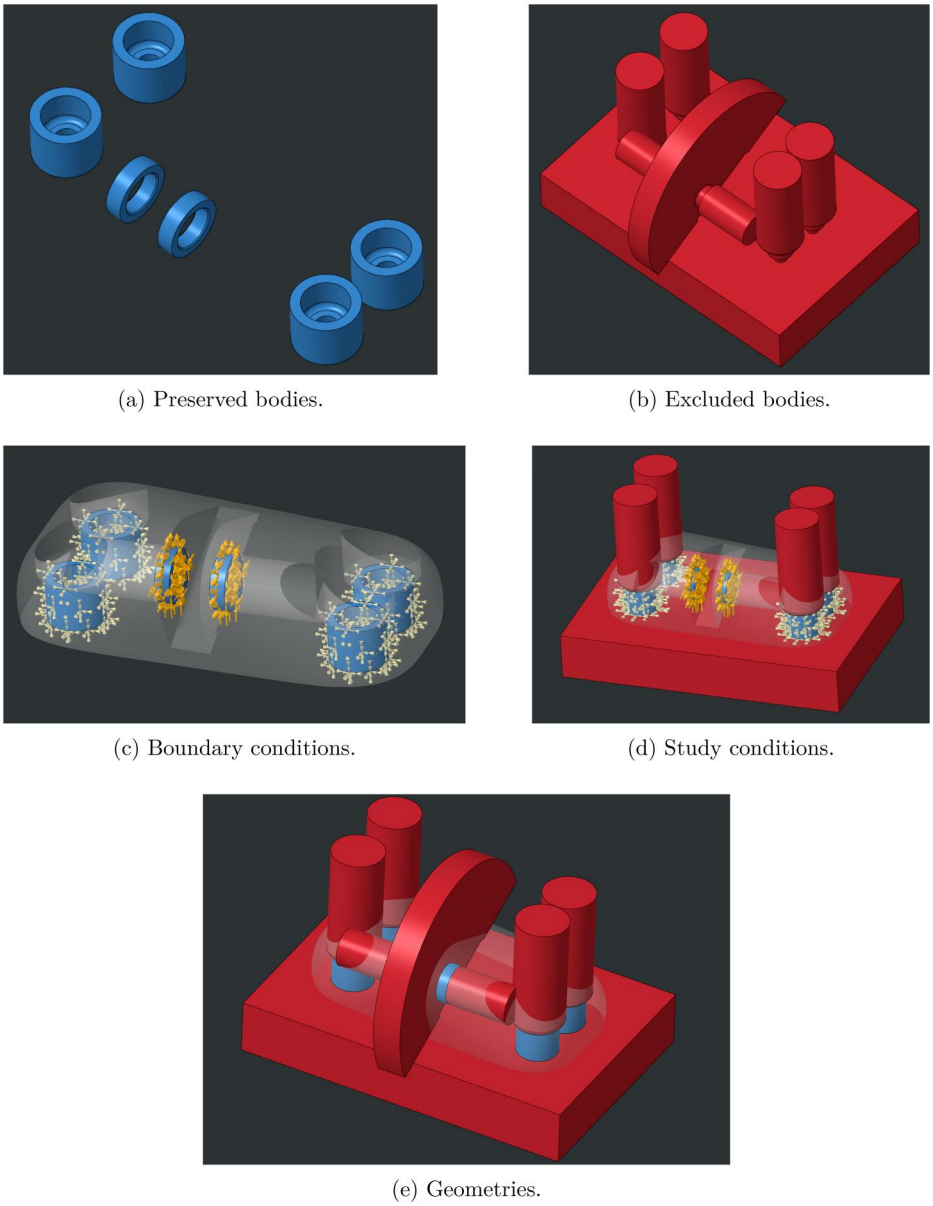
region responsible for receiving the loads. After applying the loads to the vertical rings (lugs) and imposing the fixation constraints on the four base points, the geometric and boundary conditions of the study were established, as illustrated in Figure 8.

After defining the geometries and applying the boundary conditions, the design criteria determining the generation of the component were established. Among the available optimization objectives, which generally aim at minimizing mass or maximizing stiffness, maximizing stiffness was chosen as the priority.

This strategy is based on the premise that significant mass reduction will be obtained in the subsequent stage, through the replacement of solid volume with lattice structures. Therefore, by prioritizing stiffness in the generation of the macroscopic topology, superior structural efficiency is ensured in the final component. As a control parameter, a target mass was defined, adjusted according to the properties of the selected material. Additionally, additive manufacturing constraints were applied to ensure production viability (printability): the deposition direction was defined on the positive Z-axis, and a maximum critical overhang angle of 45° was set, aiming to ensure the dimensional and surface quality of the part.

These constraints were specifically selected to align with the limitations of Laser Powder Bed Fusion (L-PBF) technology, the additive manufacturing process chosen for processing the high-performance metal alloys (AlSi10Mg and Ti6Al4V) used in this study. The 45-degree critical angle is a standard manufacturing threshold for L-PBF; by enforcing this limit during the generative phase, the algorithm is forced to create self-supporting geometries. This ensures that the final topology is not only structurally optimized but also manufacturable with minimal need for external supports, which is crucial for reducing post-processing costs and maintaining surface integrity.

The study settings defining the fidelity of the optimization were also defined, i.e. the mesh resolution and the computational cost for the generative generation, influencing the balance between geometric quality of the final part and the computational time for calculating the geometry. For maximum quality, value 10, the part will have a very smooth



(a) Preserved bodies.

(b) Excluded bodies.

(c) Boundary conditions.

(d) Study conditions.

(e) Geometries.

Figure 8. Generative part geometric conditions.

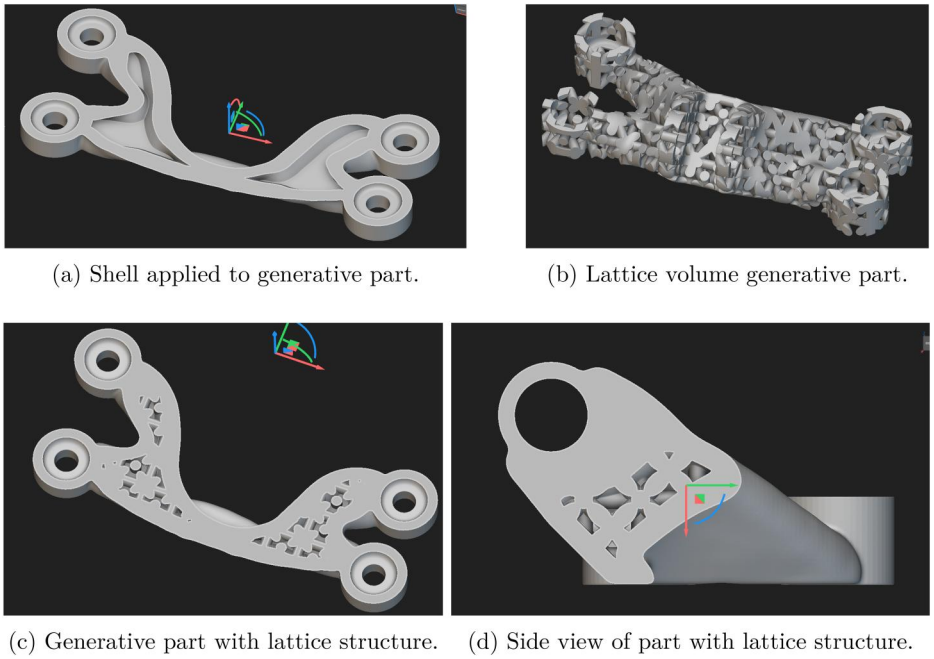
and precise geometry, capturing small details and complex curves. The higher number of iterations ensures that the part has reached convergence, being as optimized as possible. With the study fully defined, we can start the optimization, which performs all calculations locally on the computer. The same conditions were used for all parts, varying only the target mass of the optimization as it varies with the material.

3.3. Lattice applied to generative design

With the definition of the optimized lattice structure and the development of the generative geometry, the integration of these elements was proceeded with, aiming to obtain a solution that combined the maximization of strength with mass reduction, meeting the central objectives of the study. For this purpose, the nTop software was used to apply the shell technique with the filling of the internal volume by the reticulated structure. In this way, it was possible to validate the performance of the optimized component through a comparative analysis between the solid version and the hybrid version (shell-infill), proving the effectiveness in mass reduction and the gain in structural performance.

To apply lattice structures within the internal region of the generative part, the optimized macro-geometry was first imported and mathematically converted into an implicit body representation. From this continuous implicit field, an external solid shell with a uniform thickness of 6 mm was generated, creating the necessary structural envelope to encase the internal filling (Figure 9).

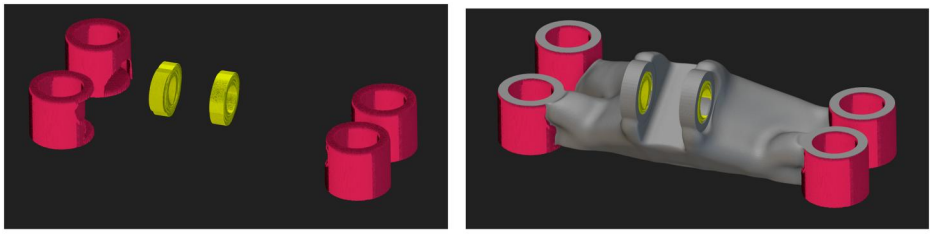
Subsequently, the internal volume bounded by the shell was designated as the domain for the lattice core. The unit cell topology and strut thickness were parameterized as variables to facilitate the iterative optimization process. This internal domain was discretized using a unit cell bounding



(a) Shell applied to generative part. (b) Lattice volume generative part.

(c) Generative part with lattice structure. (d) Side view of part with lattice structure.

Figure 9. Stages of applying lattice structure to the generative part.



(a) Boundary regions (b) Part boundary regions

Figure 10. Visualization of boundary conditions on the generative part.

box of 15 × 15 × 15 mm, with the lattice struts strictly clipped at the interface boundaries to conform precisely to the internal cavity of the geometry. The final consolidation of the hybrid component was achieved through Boolean union and intersection operations between the shell and the reticulated core, ensuring absolute material continuity and the elimination of geometric inconsistencies at the transition zones.

The configuration of the static simulation of the final component followed the methodological procedure previously validated for the lattice specimens. The discretization process began with the generation of the surface mesh from the implicit body, subsequently subjecting it to refinement stages (Remesh) to ensure element quality and result precision in regions of complex geometry. The refined surface mesh served as the domain for generating the volumetric mesh (Volume Mesh), which, associated with the mechanical properties of the selected material, constituted the definitive finite element model.

Regarding the boundary conditions, the analysis fully replicated the loading and fixation scenario defined in the *GE Bracket* case study. This included the application of fixed support constraints at the four base points and the imposition of loads on the lugs, ensuring consistency between the premises adopted in the generative design stage and the structural validation of the hybrid component (Figure 10).

To facilitate the understanding of the proposed workflow, Figure 11 presents a schematic block diagram summarizing the optimization methodology. The process is organized into two parallel streams that converge to form the final solution. The first stream (left branch) focuses on the Statistical Lattice Selection, where different cell topologies are subjected to FEA simulations and Analysis of Variance (ANOVA) to determine the optimal configuration (e.g. Gyroid in AlSi10Mg). Simultaneously, the second stream (right branch) executes the Generative Design of the component's macro-geometry, strictly adhering to L-PBF manufacturing constraints such as overhang angles. These two streams merge in the hybridization phase, where the optimized lattice structure fills the generative shell, resulting in a final hybrid component that undergoes rigorous structural validation against the design requirements.

4. Results and discussion

4.1. Numerical simulations of lattice structures

This section presents and discusses the data obtained through the Design of Experiments (DoE) numerical simulations. The central objective is to map the mechanical behavior of lattice structures as a function of design variables,

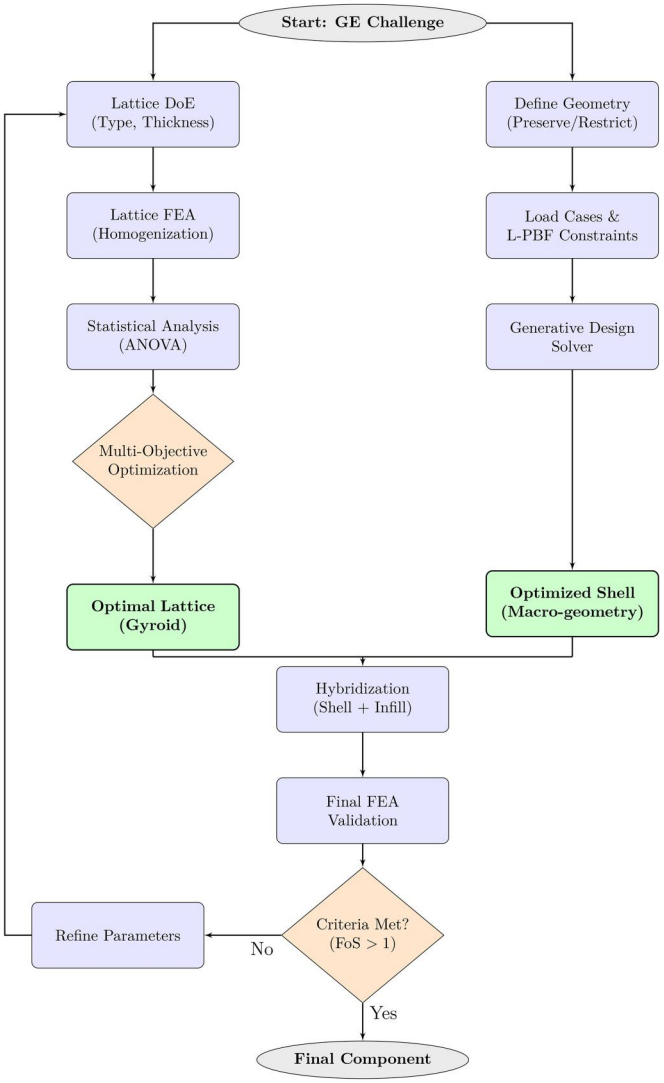


Figure 11. Flowchart of the optimization methodology: integration of Statistical Lattice selection and Generative Design.

creating a solid quantitative foundation for the optimization stages. In Table 6 the raw structural and physical performance values obtained are exposed, which underpin the sensitivity analysis and the construction of the regression models presented subsequently.

Based on the parameters established in the experimental design, simulations were performed for the proposed combinations, totaling the analysis of 15 distinct specimens. The numerical results obtained, specifically the maximum displacement and maximum Von Mises stress values, are detailed in Table 7. This dataset provided the necessary quantitative basis for the execution of the ANOVA statistical analysis and determination of response surfaces. Consequently, it was possible to proceed with mono- and multi-objective optimizations, aiming to identify the geometric and material configuration that presented the most optimized performance within the established design criteria.

The cost of each part was determined linearly and proportionally to the component's mass, using the cost per kilogram of the material. To determine the value of each variant, the component mass obtained from nTop was multiplied by the manufacturing value established for each specific material, as presented in Table 2.

4.2. Sensitivity analysis and model regression

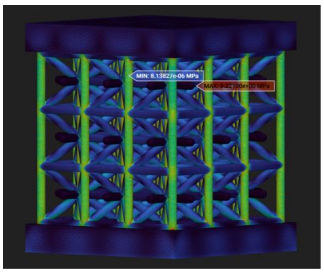
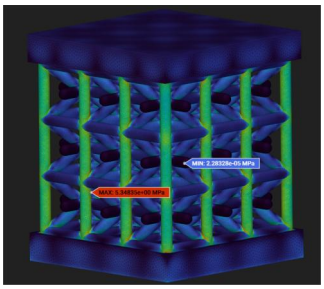
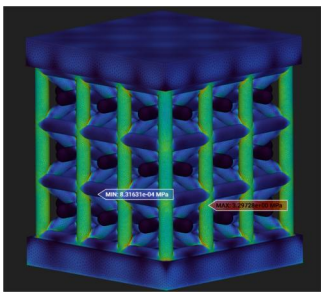
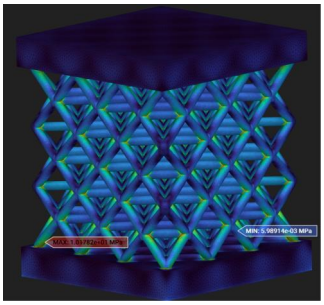
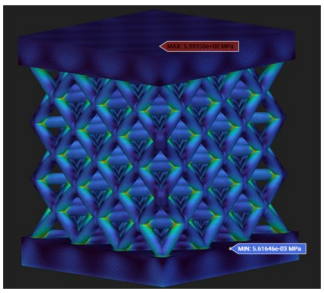
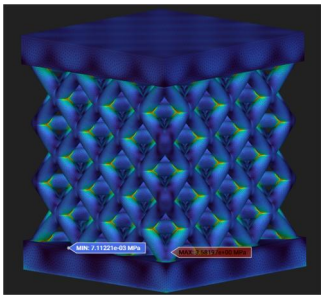
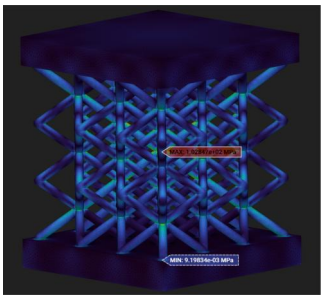
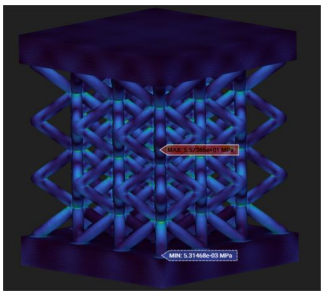
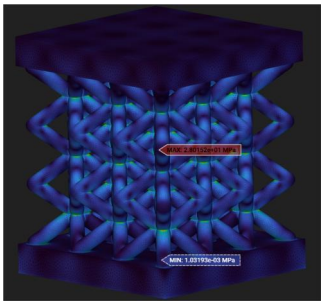
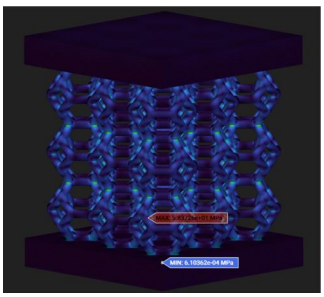
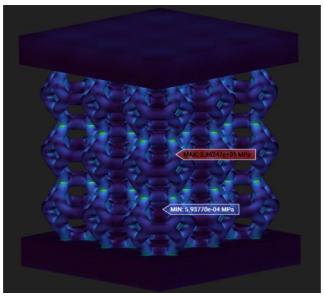
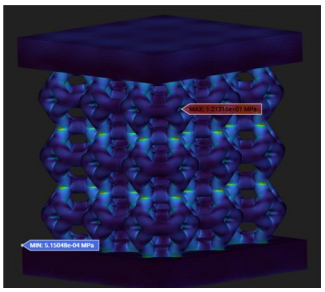
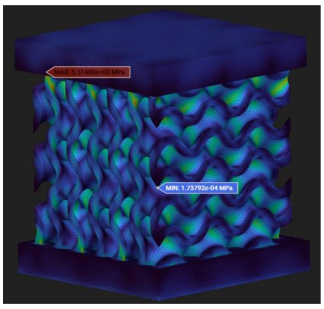
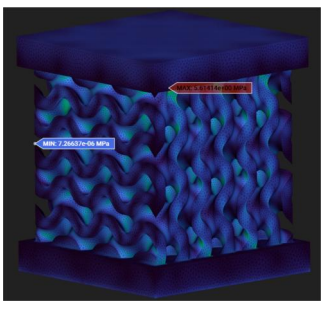
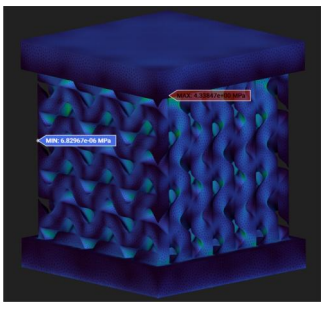
Sensitivity analysis was conducted with the purpose of identifying which design parameters exert the greatest influence on the mechanical and physical performance of the lattice structures. For this, statistical tools such as main effects plots, Pareto charts, and Analysis of Variance (ANOVA) were used, based on the results obtained by the computational simulations (Figure 12).

The analysis of the main effects reveals distinct behavioral trends depending on the chosen topology and scale. For physical and economic responses, Mass and Cost are heavily dependent on both the cell topology—with the Gyroid being the heaviest and most expensive, and the BCC the lightest and cheapest—and the material selection. Furthermore, increasing the beam diameter raises these responses in a predictable, almost linear fashion. On the mechanical side, Displacement and Stress are massively influenced by the Cell Type. Specifically, the BCC structure shows drastically higher displacement and stress peaks compared to the others, indicating lower relative stiffness, while the Gyroid and Isotruss topologies present highly efficient load distribution (lower stress peaks and displacements). Consequently, the Factor of Safety (FoS) peaks significantly for the Gyroid topology, showcasing its superior structural integrity, and naturally increases across all topologies as the strut diameter thickens.

To quantify the statistical relevance of each factor, the Pareto Chart (Figure 13) and ANOVA p-values (Table 8) were evaluated. The Pareto chart establishes a reference line for statistical significance with $\alpha = 0.05$ (95% confidence). The standardized effects confirm that Diameter (factor A) and Cell Type (factor B) govern the lattice performance. Interestingly, for Maximum Displacement, the Cell Type is the most dominant factor, tightly followed by its two-way interaction with Diameter (AB). For Mass, FoS, and Cost, the Diameter assumes the leading role, heavily followed by the interaction between Diameter and Material (AC), reflecting how density and specific cost scale with volume. The p-values confirm that Cell Type and Diameter are statistically significant ($p < 0.001$) across all evaluated metrics. Material significantly impacts Mass, Cost, FoS, and Stress/Mass ratio ($p < 0.001$), while interactions such as Cell Type × Material proved relevant, demonstrating that optimal structural performance relies heavily on the synergistic combination of geometry and the chosen alloy.

Finally, the quality of the predictive models generated from the simulated data is exceptionally high, as evidenced in Table 9. The coefficient of determination (R^2) for Mass, Relative Density (ρ^*), and Cost exceeds 99.9%, indicating an almost perfect fit dictated by geometric and volumetric rules. Crucially, the mechanical models also exhibit outstanding precision: Maximum Stress reached an R^2 of 99.37%, Factor of Safety achieved 98.97%, and Maximum Displacement demonstrated a robust 91.39% fit. These high values prove that the regression models successfully captured the structural behavior

Table 6. Visual comparison of lattice structures (material: 316 L).

Structure	Thickness 3 mm	Thickness 4 mm	Thickness 5 mm
Isotruss			
Octet			
BCC			
Kelvin			
Gyroid			

across the design space, encompassing the non-linearities (such as the quadratic effect of the diameter) and the complex two-way interactions that define lattice performance.

Furthermore, despite the exceptionally high coefficient of determination (R^2) values across both geometric and mechanical responses indicating a highly reliable statistical fit, the predictive capability of these models must be interpreted with caution when applied to real-world scenarios. The statistical equations presented here are derived from continuum mechanics simulations which assume ideal material homogeneity and perfect geometric fidelity. In practical Additive Manufacturing applications, however, the mechanical behavior is significantly influenced by process-inherent defects such as microporosity, surface roughness, and anisotropy

Table 7. DoE Experimental results for different cell types, diameters, and materials.

Cell	ø (mm)	Mat	Disp (mm)	Stress (MPa)	Mass (g)	Str/Mass (MPa/g)	FoS (UTS)	ρ^*	Cost ($)
Isotruss	3	316L	0.001390	9.221	1115.32	0.00827	64.53	0.45	613.43
Octet	3	316L	0.001589	10.178	1206.66	0.00843	58.46	0.49	663.66
BCC	3	316L	0.018437	102.847	927.29	0.11091	5.79	0.38	510.01
Kelvin	3	316L	0.005294	58.373	988.82	0.05903	10.19	0.40	543.85
Gyroid	3	316L	0.000530	3.661	1498.38	0.00244	162.52	0.61	824.11
Isotruss	4	316L	0.000732	5.348	1359.68	0.00393	111.26	0.55	747.82
Octet	4	316L	0.000749	5.993	1497.03	0.00400	99.28	0.61	823.37
BCC	4	316L	0.007250	55.736	1058.40	0.05266	10.68	0.43	582.12
Kelvin	4	316L	0.002010	24.625	1153.86	0.02134	24.16	0.47	634.62
Gyroid	4	316L	0.000380	2.768	1724.65	0.00160	214.96	0.70	948.56
Isotruss	5	316L	0.000434	3.297	1637.68	0.00201	180.47	0.66	900.72
Octet	5	316L	0.000397	3.582	1810.03	0.00198	166.11	0.73	995.52
BCC	5	316L	0.003227	28.152	1215.00	0.02317	21.14	0.49	668.25
Kelvin	5	316L	0.001039	12.132	1344.05	0.00903	49.04	0.54	739.23
Gyroid	5	316L	0.000295	2.449	1930.23	0.00127	242.96	0.78	1061.63
Isotruss	3	AlSi10Mg	0.003972	9.172	370.84	0.02473	83.95	0.46	285.55
Octet	3	AlSi10Mg	0.004534	10.141	401.28	0.02527	75.93	0.50	308.98
BCC	3	AlSi10Mg	0.052185	100.443	308.23	0.32587	7.67	0.38	237.34
Kelvin	3	AlSi10Mg	0.001502	57.582	328.78	0.17514	13.37	0.41	253.16
Gyroid	3	AlSi10Mg	0.000150	3.621	498.21	0.00727	212.65	0.62	383.62
Isotruss	4	AlSi10Mg	0.002086	5.288	452.09	0.01170	145.61	0.56	348.11
Octet	4	AlSi10Mg	0.002137	5.947	497.76	0.01195	129.48	0.62	383.28
BCC	4	AlSi10Mg	0.020461	54.415	351.92	0.15462	14.15	0.44	270.98
Kelvin	4	AlSi10Mg	0.005689	24.348	383.66	0.06346	31.62	0.48	295.42
Gyroid	4	AlSi10Mg	0.001080	2.749	573.44	0.00479	280.10	0.71	441.55
Isotruss	5	AlSi10Mg	0.001236	3.281	544.53	0.00603	234.68	0.68	419.29
Octet	5	AlSi10Mg	0.001131	3.541	601.83	0.00588	217.46	0.75	463.41
BCC	5	AlSi10Mg	0.009089	27.397	403.99	0.06782	28.11	0.50	311.07
Kelvin	5	AlSi10Mg	0.002941	11.937	446.90	0.02671	64.50	0.55	344.11
Gyroid	5	AlSi10Mg	0.000840	2.443	641.80	0.00381	315.19	0.80	494.19
Isotruss	3	Ti6Al4V	0.002380	9.184	617.61	0.01487	228.66	0.45	1296.98
Octet	3	Ti6Al4V	0.002720	10.154	668.19	0.01520	206.82	0.49	1403.20
BCC	3	Ti6Al4V	0.031300	101.460	513.49	0.19759	20.70	0.38	1078.33
Kelvin	3	Ti6Al4V	0.009010	57.840	547.56	0.10563	36.31	0.40	1149.88
Gyroid	3	Ti6Al4V	0.000920	3.674	829.73	0.00443	571.58	0.61	1742.43
Isotruss	4	Ti6Al4V	0.001250	5.308	752.92	0.00705	395.63	0.55	1581.13
Octet	4	Ti6Al4V	0.001280	5.962	828.98	0.00719	352.23	0.61	1740.86
BCC	4	Ti6Al4V	0.012292	54.855	586.09	0.09360	38.28	0.43	1230.78
Kelvin	4	Ti6Al4V	0.003410	24.440	638.95	0.03825	85.92	0.47	1341.80
Gyroid	4	Ti6Al4V	0.000650	2.774	955.02	0.00290	757.03	0.70	2005.54
Isotruss	5	Ti6Al4V	0.000742	3.287	906.87	0.00362	638.88	0.66	1904.43
Octet	5	Ti6Al4V	0.000677	3.555	1002.30	0.00355	590.72	0.73	2104.84
BCC	5	Ti6Al4V	0.005464	27.760	672.81	0.04126	75.65	0.49	1412.90
Kelvin	5	Ti6Al4V	0.001760	12.000	744.27	0.01612	175.00	0.54	1562.97
Gyroid	5	Ti6Al4V	0.000506	2.451	1068.87	0.00229	856.79	0.78	2244.62

induced by layer-by-layer deposition, which are not captured in the simplified regression models. Consequently, while these statistical tools are highly effective for the comparative ranking of topologies and identifying the 'sweet spot' for design exploration (as demonstrated by the selection of the most efficient structures) they act as a proxy for theoretical performance. Absolute failure prediction in a physical aeronautical or biomedical environment would require calibrating these models with experimental data to account for the stiffness degradation and stress concentrations caused by manufacturing deviations.

4.3. Optimization

Following the statistical analysis of the data, the optimization stage was proceeded with to determine the ideal design parameters. This stage was divided into two approaches: mono-objective optimization, focused on maximizing or minimizing a single response in isolation, and multi-objective optimization, which seeks the best compromise between conflicting responses. Prior to the execution of the optimization algorithms, the correlation matrix of the responses, illustrated in Figure 14, was analyzed to understand the interdependencies between the design objectives. The matrix revealed a strong positive correlation between mass and cost, indicating that mass minimization directly entails the reduction of manufacturing costs. In contrast, the inherent conflict between mass reduction and structural integrity (represented by stress and Factor of Safety) was observed, where material removal tends to increase the acting stresses.

4.3.1. Mono-objective optimization

In this phase, the Genetic Algorithm (GA) was used to find the optimal values for individual objective functions. As presented in Table 10, when the objective is purely the reduction of mass, cost minimization, or maximizing the specific

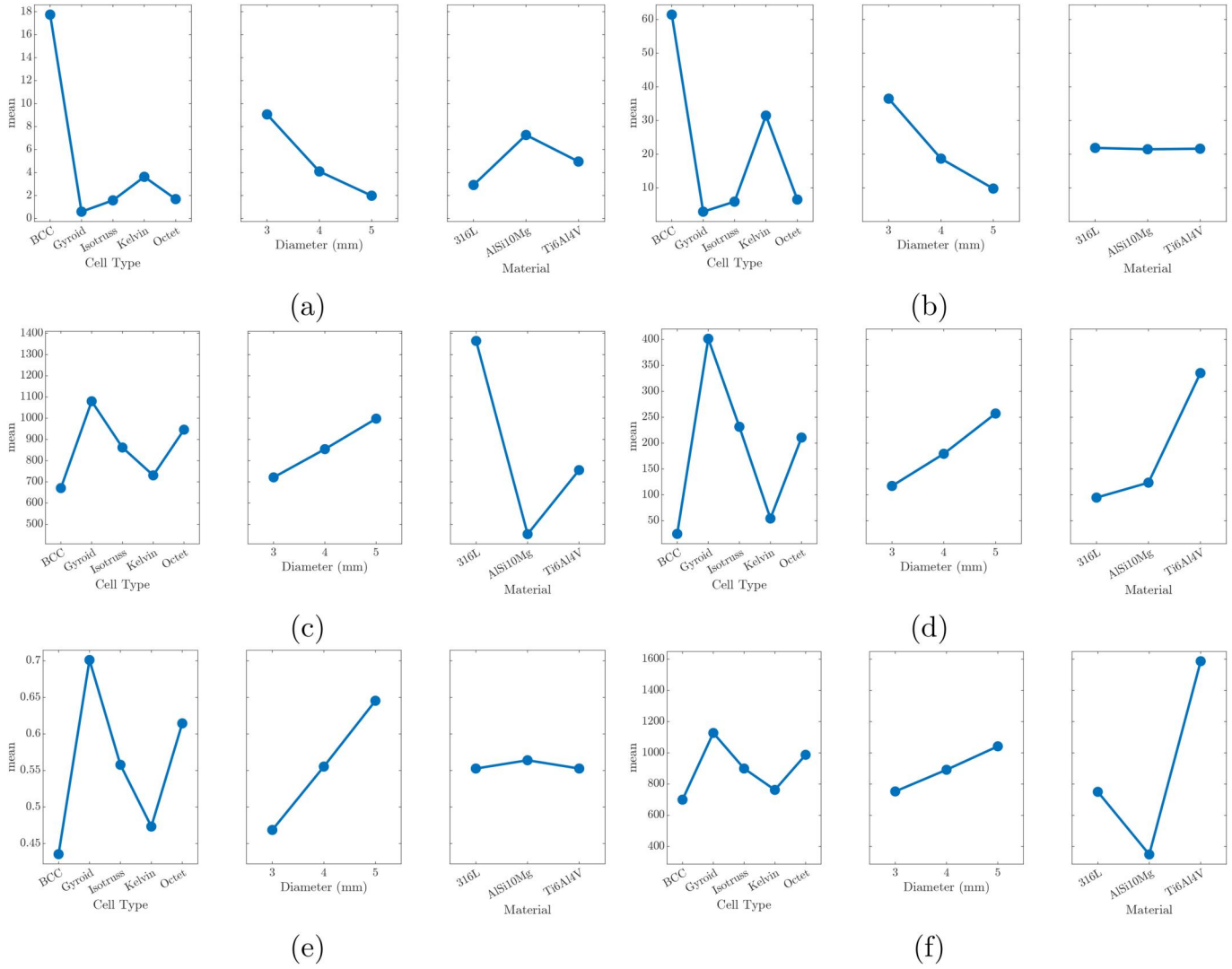


Figure 12. Main effects: (a) displacement, (b) max stress, (c) mass, (d) FoS, (e) ρ^*, (f) cost.

strength (Stress/Mass), the BCC structure with a 3 mm diameter manufactured in AlSi10Mg proved to be the best candidate, due to its highly porous geometry and lower material volume. Similarly, for minimizing the relative density (ρ^*), the BCC topology at 3 mm also excels, though the algorithm selected the 316 L alloy for this specific metric.

On the other hand, when the objective shifts toward mechanical integrity, such as minimizing maximum stress and displacement, or maximizing the Factor of Safety (FoS), different optimal configurations emerge. To minimize maximum displacement, the Isotruss structure manufactured in 316 L with a diameter of approximately 4.96 mm was the optimal choice. For minimizing maximum stress, the Gyroid structure in Ti6Al4V at roughly 3.96 mm was superior. Notably, to achieve the absolute maximum FoS (863.8), the GA selected the Gyroid cell manufactured in Ti6Al4V with the maximum allowable diameter (5.00 mm). It is worth mentioning that for the minimization of maximum stress and displacement, the regression model predicted extremely small positive values (approaching zero). Mathematically, the GA converged to the lowest possible roots of the continuous regression equations, which physically translates to configurations of extreme stiffness and practically negligible deformation under the evaluated boundary conditions.

This isolated analysis is fundamental to understanding the extreme performance limits of each topology and material combination. However, it proves insufficient for the final definition of an efficient component, given that a part optimized solely for lightness (such as the 3 mm BCC) may lack the required structural robustness, while one optimized exclusively for maximum safety (like the dense 5 mm Gyroid in Titanium) will inevitably become excessively heavy and economically unviable.

4.3.2. Multi-objective optimization

Multi-objective optimization was conducted to find a compromise solution (Pareto front) balancing the conflicting variables of maximum stress (σ_{max}), mass, and relative density (ρ^*). After the initial generation and filtering phases, 300 valid solutions were identified. To systematically rank these solutions and select the most appropriate configuration, the TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) method was employed. Since TOPSIS relies

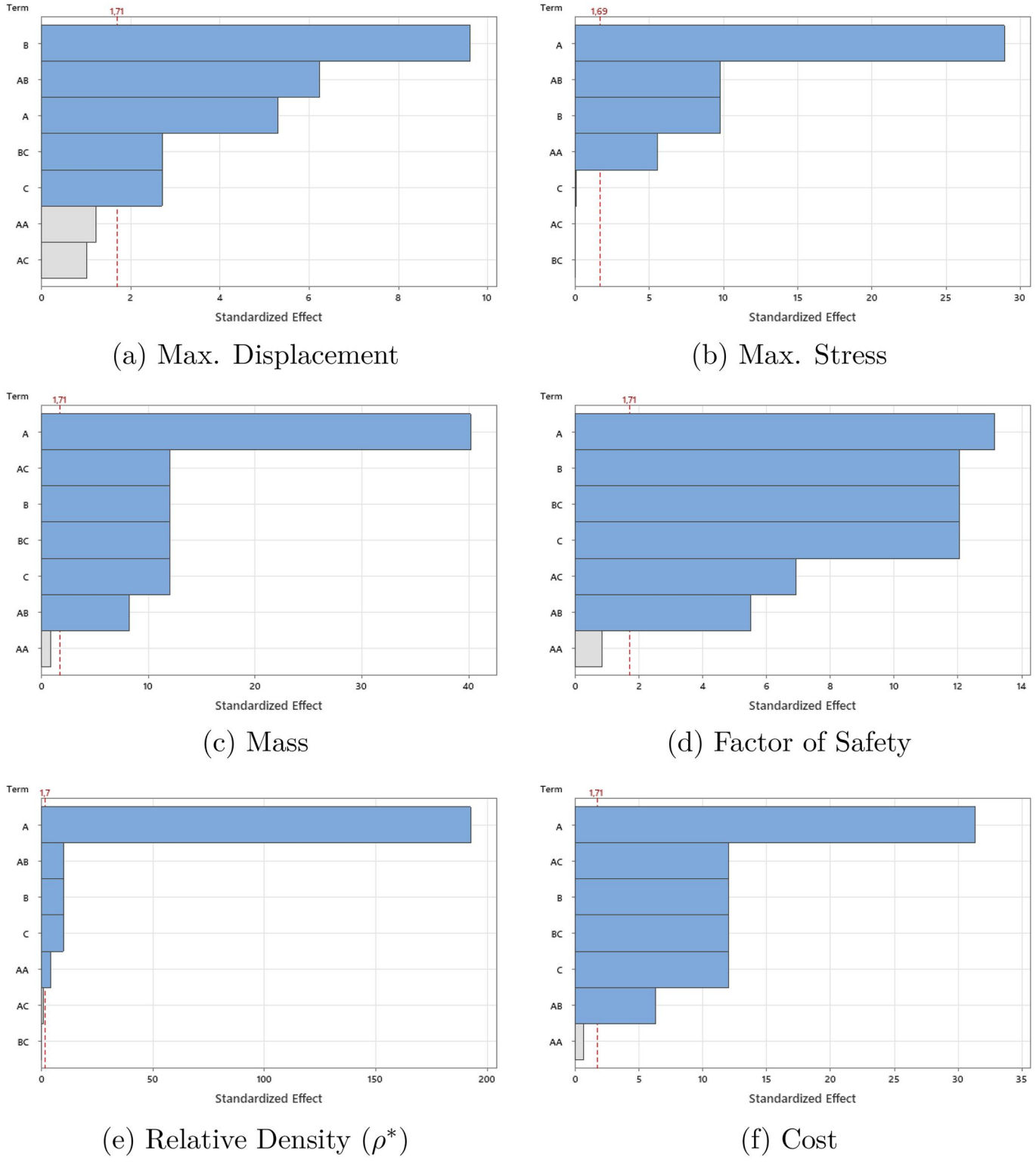


Figure 13. Pareto effects. Effects exceeding the red reference line are statistically significant, $p < 0.05$ (legend: a = cell type, B = diameter, C = material). A gray bar represents a term not in the model.

Table 8. Summary of p-values for the analysis of variance (ANOVA) of responses.

Parameter	Max. Disp	Max. Stress	Mass	Str/Mass	FoS	ρ^*	Cost
Cell Type	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Diameter	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Material	0.012	–	0.000	0.000	0.000	0.000	0.000
Diameter × Cell Type	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Diameter × Material	–	–	0.000	0.005	0.000	–	0.000
Cell Type × Material	0.012	–	0.000	0.001	0.000	–	0.000

Note: "-" indicates that the term was not included in the final model for this response.

Table 9. Summary of regression model fit.

Response	R^2 (%)	R^2_{adj} (%)	R^2_{pred} (%)
Max. Displacement	91.39	84.85	69.01
Max. Stress	99.37	99.18	98.99
Mass	99.90	99.81	99.50
Stress/Mass	96.03	92.06	79.80
FoS	98.97	98.03	95.03
ρ^*	99.97	99.96	99.94
Cost ($)	99.90	99.80	99.47

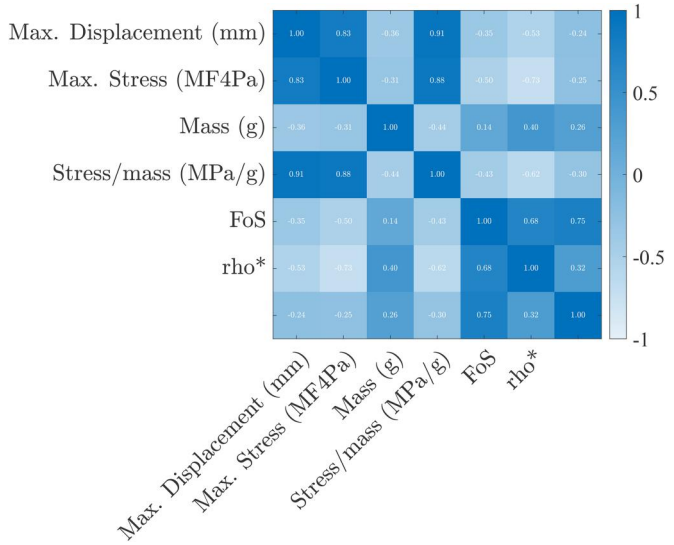


Figure 14. Response correlation matrix.

Table 10. Summary of Mono-objective optimization results (GA).

Objective	Cell	Material	Diam (mm)	Predicted (Value)
Min Max. Displacement (mm)	Isotruss	316 L	4.96	$7.47(10^{-11})$
Min Max. Stress (MPa)	Gyroid	Ti6Al4V	3.96	$4.48(10^{-8})$
Min Mass (g)	BCC	AlSi10Mg	3.00	329.44
Max Stress/Mass (MPa/g)	BCC	AlSi10Mg	3.00	0.2861
Max FoS (-)	Gyroid	Ti6Al4V	5.00	863.80
Min ρ^* (-)	BCC	316 L	3.00	0.3740
Min Cost ($)	BCC	AlSi10Mg	3.00	266.70

heavily on the relative importance assigned to each criterion, three different objective weighting strategies were evaluated to thoroughly explore the decision space: Shannon Entropy, CRITIC, and Equal Weights.

Each weighting method employs a distinct mathematical approach to assign importance to the objective functions, directly influencing the final topological selection. Let n be the total number of valid alternatives ($n = 300$) and m be the number of objective criteria ($m = 3$). For a given performance matrix where f_{ij} represents the value of the i-th alternative for the j-th criterion:

- **Shannon Entropy:** This objective weighting method evaluates the degree of dispersion and data diversity within the evaluation matrix. First, the objective values are normalized into probability distributions p_{ij} to eliminate physical unit dimensions:

$$p_{ij} = \frac{f_{ij}}{\sum_{i=1}^{n} f_{ij}} \tag{6}$$

Next, the information entropy E_j for each criterion j is calculated as:

$$E_j = -\frac{1}{\ln(n)} \sum_{i=1}^{n} p_{ij} \ln(p_{ij}) \tag{7}$$

Criteria with greater variance (lower entropy) provide more discriminatory information. The final weight w_j is determined by the degree of divergence $(1 - E_j)$:

$$w_j = \frac{1 - E_j}{\sum_{j=1}^{m} (1 - E_j)} \tag{8}$$

In this analysis, the Shannon method assigned weights of 0.565 for σ_{max}, 0.432 for Mass, and 0.003 for ρ^*. As a result, this strategy heavily prioritized mechanical strength and mass reduction, practically nullifying the influence of relative density in the decision-making process.

- **CRITIC (Criteria Importance Through Intercriteria Correlation):** The CRITIC method offers a more comprehensive evaluation by accounting for both the contrast intensity of each criterion and the degree of conflict between them. The objectives are first normalized using a min-max scaling:

$$\overline{f}_{ij} = \frac{f_{ij} - f_j^{\min}}{f_j^{\max} - f_j^{\min}} \tag{9}$$

The method then calculates the standard deviation σ_j of each normalized criterion and the linear correlation coefficient r_{jk} between criteria j and k. The amount of information C_j contained in the j-th criterion, which represents the conflict, is given by:

$$C_j = \sigma_j \sum_{k=1}^{m} (1 - r_{jk}) \tag{10}$$

The final objective weight is obtained by normalizing these information measures:

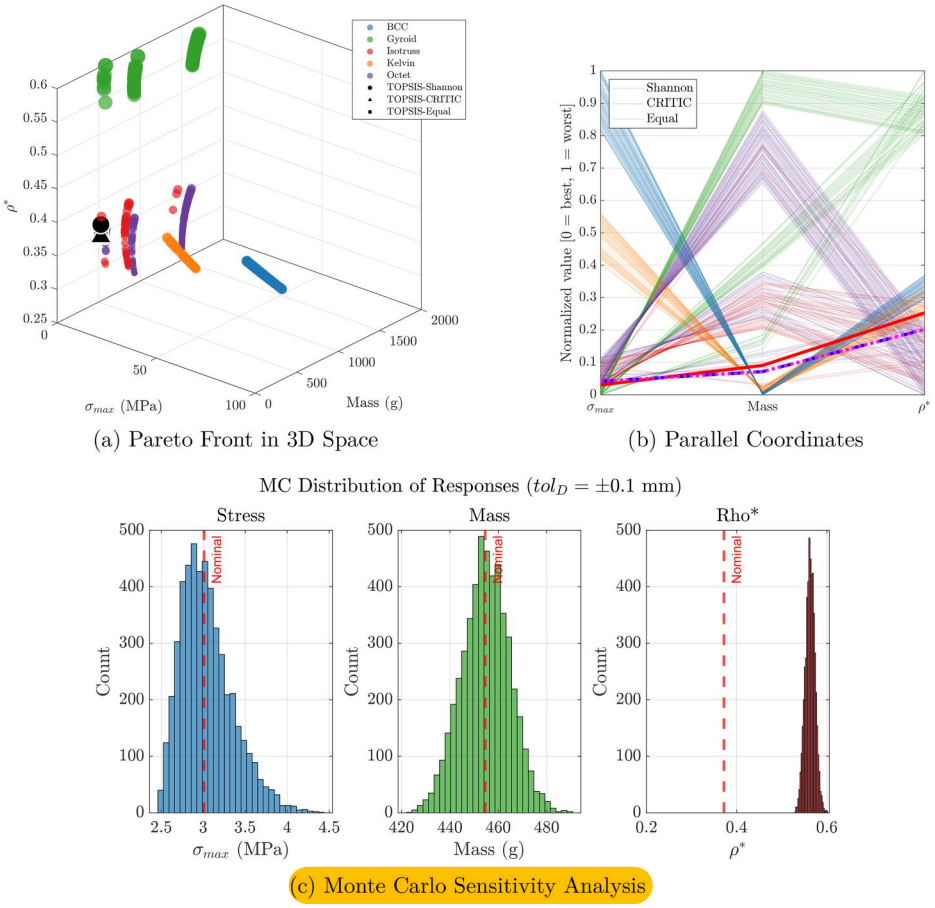
$$w_j = \frac{C_j}{\sum_{j=1}^{m} C_j} \tag{11}$$

This approach yielded a balanced distribution of weights: 0.408 for σ_{max}, 0.359 for Mass, and 0.233 for ρ^*, properly acknowledging the relative density while maintaining a slight preference for structural integrity and lightness.

- **Equal Weights:** This straightforward formulation assumes that all design criteria are strictly equally critical to the overall success of the component. It assigns a uniform weight to each criterion:

$$w_j = \frac{1}{m} \tag{12}$$

Given $m = 3$, it assigns a uniform weight of 0.333 to maximum stress, mass, and relative density, ensuring an unbiased selection that seeks the true geometric "sweet spot" without mathematically favoring any specific extreme.



(a) Pareto Front in 3D Space

(b) Parallel Coordinates

(c) Monte Carlo Sensitivity Analysis

Figure 15. Multi-objective optimization graphical results displaying the pareto front, parallel coordinates, and Monte Carlo robustness analysis.

Table 11. Optimal solutions obtained by the TOPSIS method for multi-objective optimization using different weighting strategies.

Optimization	Weights Method	Cell [-]	Material [-]	Diam [mm]	Max Stress [MPa]	Mass [g]	ρ^* [-]
minimize $\mathcal{F}_1(\mathbf{X}) =$ Stress	Shannon ($C_i = 0.9522$)	Isotruss	AlSi10Mg	3.9873	3.0103	454.5607	0.3717
minimize $\mathcal{F}_2(\mathbf{X}) =$ Mass	CRITIC ($C_i = 0.9410$)	Isotruss	AlSi10Mg	3.7202	4.1475	428.1106	0.3563
minimize $\mathcal{F}_3(\mathbf{X}) = \rho^*$	Equal ($C_i = 0.9274$)	Gyroid	AlSi10Mg	3.7202	4.1475	428.1106	0.3563

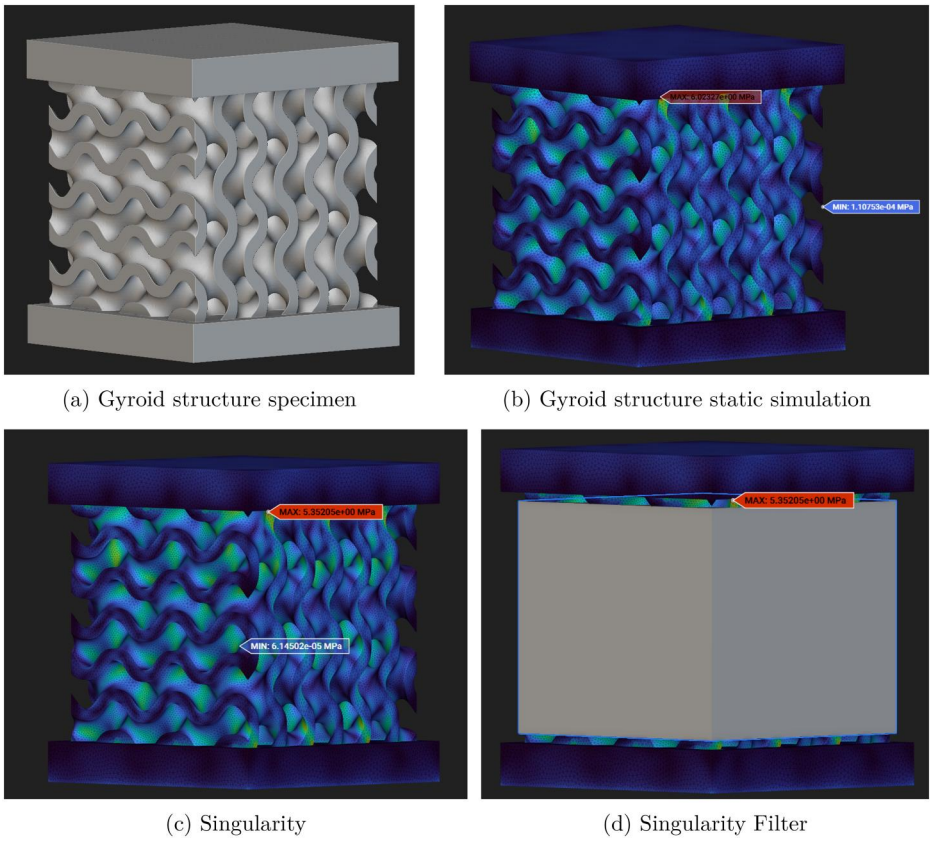
The tradeoffs between the objectives are visualized in the parallel coordinates plot (Figure 15) and the 3D Pareto front (Figure 15). The Shannon weighting heavily prioritized stress reduction, resulting in an Isotruss cell with 3.9873 mm diameter. The CRITIC method provided a more balanced distribution but still favored the Isotruss topology (3.7202 mm). However, considering the overall project goal of finding a "sweet spot" without bias toward a single extreme, the Equal weights strategy (0.333 for each objective) was selected.

Under the Equal weights criterion, the global optimal configuration is the gyroid cell, manufactured in AlSi10Mg with a diameter of 3.7202 mm, yielding a maximum stress of 4.1475 MPa, a mass of 428.1106 g, and a relative density (ρ^*) of 0.3563, as detailed in Table 11. The Gyroid, being a Triply Periodic Minimal Surface (TPMS), offers a continuous geometry without sharp nodes, allowing for a more uniform distribution of stresses compared to strut-based lattices like BCC or Isotruss. Furthermore, a Monte Carlo sensitivity analysis assuming a manufacturing tolerance of ± 0.1 mm demonstrates the robustness of this optimal solution, with the stress and mass distributions remaining well-behaved around the nominal values (Figure 15).

To address potential stress concentrations at the interface between the lattice core and the solid plates (a common point of structural failure) a singularity filter was implemented in the refined models. This geometric smoothing ensures a more uniform stress distribution at the connection points. Figure 16 illustrates the specimen details, the static simulation results, and the effectiveness of the filtering process in eliminating stress singularities at the interfaces.

4.4. Generative lattice design

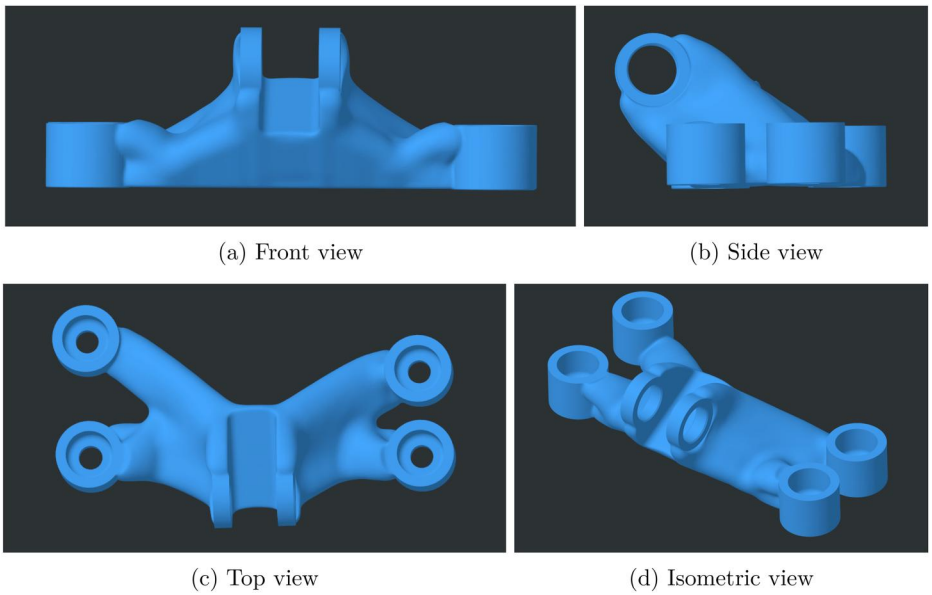
After defining the geometries and study configurations, the process of optimization and component generation *via* Generative Design began. Based on the results obtained in the cell selection stage, the AlSi10Mg aluminum alloy was identified as the best-performing material. Therefore, this material was adopted, according to the properties described in Table 1, for the generation of the base geometry and subsequent filling with the internal lattice structure.



(a) Gyroid structure specimen (b) Gyroid structure static simulation

(c) Singularity (d) Singularity Filter

Figure 16. Gyroid lattice mesh details and singularity filtering.



(a) Front view (b) Side view

(c) Top view (d) Isometric view

Figure 17. Views of the generative part and its variations.

In the analysis of the resulting geometry, it is observed that the volumes configured as preserved were fully maintained in the final part, while the exclusion regions remained free of material, ensuring the correct application and assembly of the component. The generated topology is characterized by organic shapes and the definition of optimized load paths, resulting in the selective removal of material in areas of low stress density (Figure 17). Once the generative geometry was obtained, the model was treated to incorporate the internal lattice structures, with the objective of promoting additional mass reduction without compromising the component's global stiffness.

With the results obtained, we filled the interior of the generative part with a 6 mm shell and a gyroid structure with 3 mm thickness, manufactured in AlSi10Mg solely to maintain fidelity to the case study proposed for the generative part.

The final stage consisted of integrating the optimized cellular structure into the geometry obtained by Generative Design. Based on the results of the multi-objective optimization, which identified the gyroid topology manufactured in Ti6Al4V as the best compromise solution between lightness and strength, the filling of the component's internal volume was proceeded with.

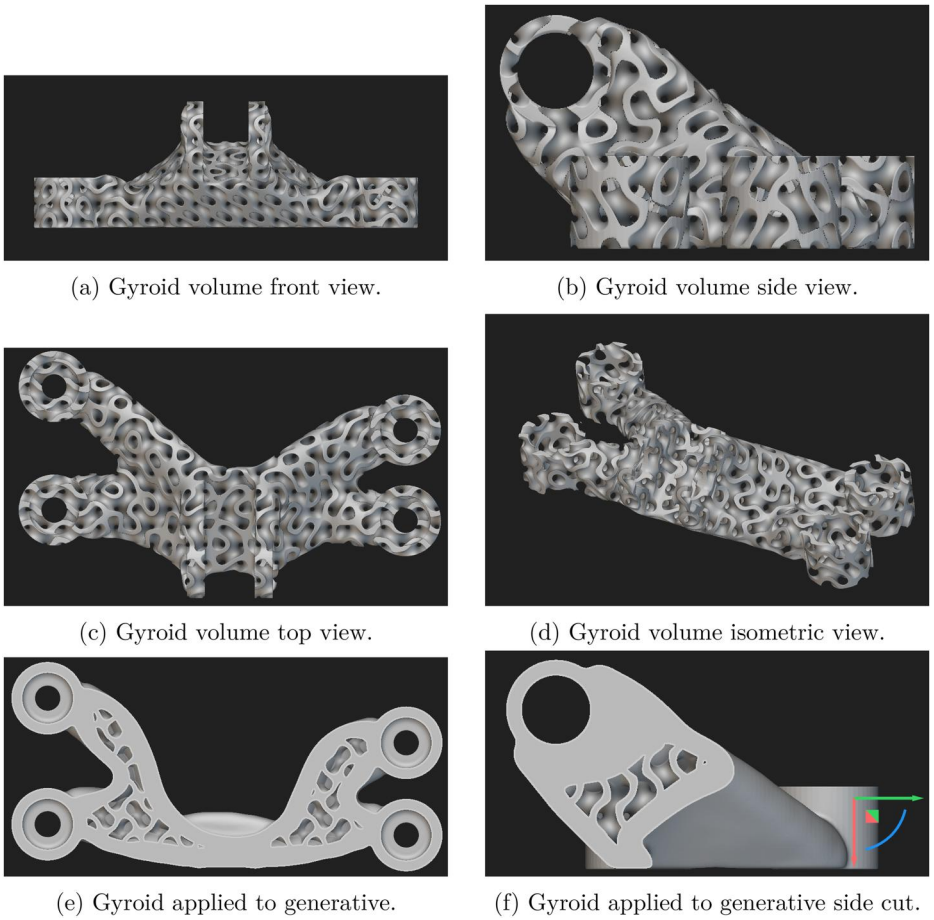
To ensure the integrity of the connection interfaces and surface resistance against external impacts, an external solid wall (shell) with a thickness of 6 mm was defined from the original geometry. The resulting internal volume was filled by the lattice gyroid mesh with a beam thickness of 3 mm. The boolean union operation between the solid shell and the lattice structure resulted in a hybrid component (Figure 18).

The performance of the application of the proposed methodology, which integrates Generative Design with Lattice Structures, is evaluated in comparison with the original component provided in the *GE Jet Engine Bracket* challenge. The comparison considers the criteria of mass, structural integrity, factor of safety, and economic viability, using the properties of the Titanium alloy Ti6Al4V. The consolidated data from the final simulations are presented in Figure 19 and Table 12.

The main objective of additive manufacturing is weight reduction due to the possibility of fabricating optimized geometries. It is observed that the Generative Design approach in the solid version achieved an expressive mass reduction of approximately 62% in relation to the original part, reducing from 2052 g to 778 g. The application of the gyroid lattice structure allowed for additional optimization, where the replacement of the solid core with the cellular mesh resulted in an extra reduction of about 70 g, decreasing the final mass to 708.61 g. This represents a total reduction of 65.5% compared to the original design. Consequently, the estimated manufacturing cost followed this linear trend. The optimized part with lattice presented an approximate cost of 1,488.00 dollars, representing theoretical savings of 2,821.00 dollars per unit relative to the original component, validating the economic viability of the methodology for low-scale production.

The structural analysis reveals the classic tradeoff between material removal and stiffness. Regarding displacement, the original part, being extremely robust, showed a minimum displacement of 0.155 mm. The optimized versions presented greater flexibility, with displacements in the range of 0.29 mm to 0.33 mm. Although displacement doubled, it still remains within an acceptable order of magnitude for many applications, demonstrating that the generative topology preserved global stiffness efficiently.

Regarding stress and the factor of safety, the results demonstrate the technical feasibility of the design. The original part operates with a Factor of Safety of 2.25, indicating considerable oversizing. The optimized parts, in turn, operate with factors of safety of 1.52 (Generative Solid) and 1.52 (Generative + Lattice). Both values are above 1.0, ensuring



(a) Gyroid volume front view.

(b) Gyroid volume side view.

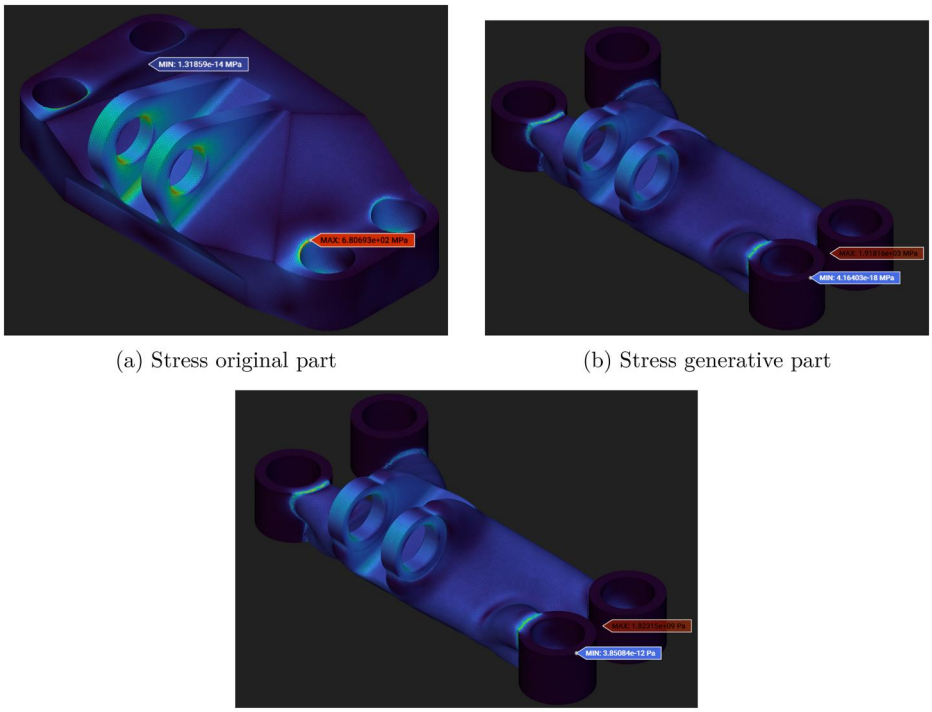
(c) Gyroid volume top view.

(d) Gyroid volume isometric view.

(e) Gyroid applied to generative.

(f) Gyroid applied to generative side cut.

Figure 18. Integration of optimal results.



(a) Stress original part

(b) Stress generative part

(c) Stress Generative part with Lattice

Figure 19. Comparative von mises stress distribution in the three configurations.

Table 12. Performance comparison: Original part vs. optimizations (Ti6Al4V).

	Part Configuration		
Parameter	Original (GE)	Generative (Solid)	Generative + Lattice
Mass [g]	2052.13	778.51	708.61
Max Stress (Von Mises) [MPa]	462.67	685.71	686.34
Max Displacement [mm]	0.155	0.292	0.327
Relative Density (ρ^*)	–	1.00	0.91
Factor of Safety (FoS)*	2.25	1.52	1.52
Estimated Cost [$]**	4,309.47	1,634.87	1,488.08

Notes: (*) Based on Ti6Al4V UTS = 1040 MPa. (**) Estimate based on mass ($2100/kg).

that the parts will not fail under design loads, while eliminating unnecessary excess material.

The maximum stress observed in the hybrid component (686 MPa) compared to the solid version (686 MPa) highlights the efficiency of the lattice structure in redistributing internal loads even with less material. Unlike solid material, where stresses tend to accumulate on external surfaces, cellular filling allows load flow to permeate the internal volume of the part, relieving the overload on the external shell and slightly raising the safety margin of the final component.

This mechanical advantage is intrinsic to the topology of Triply Periodic Minimal Surfaces (TPMS), the category to which the gyroid belongs. Its geometry is characterized by zero mean curvature and the absence of nodes or sharp intersections—common elements in strut-based lattices that act as stress concentrators. This morphological continuity promotes more uniform and efficient energy dissipation under loading, increasing fracture toughness and fatigue resistance of the assembly. Such attributes make the application of gyroid cores advantageous for various engineering sectors where the capacity to absorb vibrations and resist dynamic loads without catastrophic failure is a critical design requirement.

The proposed methodology was extremely effective in mass and cost reduction, surpassing weight reduction goals. However, the stress results indicate that the generative geometry generated was excessively aggressive in material removal for the extreme loads of the GE challenge, especially considering torque and combined loads. For a safe practical application, a design refinement would be necessary, such as increasing the shell thickness or increasing the relative density of the internal lattice to raise the Factor of Safety above 1.0, while still maintaining a significant weight advantage over the original component.

5. Conclusion and recommendations

The present work achieved its main objective by proposing and validating a methodology for the integration of optimized lattice structures into components developed *via* Generative Design, aiming at Additive Manufacturing applications. Through a systematic approach involving computational simulation, statistical analysis, and optimization algorithms, it was possible to map the mechanical behavior

of different cellular topologies and identify the most efficient structural configurations.

Numerical and statistical analyses demonstrated that geometric parameters exert a decisive influence on structural performance. The Analysis of Variance evidenced that Cell Type and Beam Diameter are the most significant factors governing stiffness and maximum stress, while Material Selection emerged as the dominant factor determining the mass and final cost of the component. Furthermore, mono-objective optimization revealed the performance limits of each structure in isolation. While the BCC cell proved superior for absolute mass reduction, the Isotruss and Octet cells presented higher raw mechanical strength, indicating that isolating these design objectives is insufficient for a balanced engineering solution.

In this context, multi-objective optimization identified the Gyroid structure, manufactured in AlSi10Mg aluminum alloy with a nominal thickness of 3 mm, as the optimal compromise solution. This configuration presented the best strength-to-weight ratio, distributing stresses uniformly due to its Triply Periodic Minimal Surface geometry, which effectively mitigates the stress concentrations commonly found in strut-based lattices with sharp nodes. The application of this optimized topology to the GE Jet Engine Bracket case study, utilizing a hybrid shell-infill technique, successfully validated the proposed methodology. The final component maintained the necessary structural integrity to withstand complex operational loads while achieving a total mass reduction of 65.5% compared to the original conventionally manufactured design—a reduction superior to what would be possible with solid topology optimization alone. Therefore, it is concluded that the hybridization of Generative Design and lattice structures, when guided by statistical optimization, constitutes a highly effective methodology for modern engineering, fully exploiting the geometric freedom offered by Additive Manufacturing.

To further advance this research, experimental validation through the physical additive manufacturing of specimens and the final component is highly recommended. Comparing experimental mechanical tests with the numerical simulations is essential to calibrate the finite element models and ensure the absolute fidelity of the virtual results. Additionally, considering the aerospace application of the GE Bracket, future studies should perform dynamic analyses to evaluate the behavior of the hybrid lattice structures under cyclic fatigue loading and verify the natural frequency of the assembly, ensuring structural reliability under engine-induced vibrations.

Author contributions

CRediT: **Igor de Magalhães Catini**: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, and Visualization. **Guilherme Ferreira Gomes**: Conceptualization, Methodology, Resources, Writing - Review & Editing, Supervision, and Project Administration. Both authors contributed to the design and implementation of the research, to the analysis of the results, and to the writing of the manuscript. Both authors reviewed and approved the final version of the manuscript to be published and agree to be accountable for all aspects of the work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG) under Grants APQ-02290-23 and APQ-00062-24.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

[1] M.B. Francisco, D.M. Junqueira, G.A. Oliver, J.L.J. Pereira, S.S. da Cunha, Jr Jr, and G.F. Gomes, Design optimizations of carbon fibre reinforced polymer isogrid lower limb prosthesis using particle swarm optimization and lichtenberg algorithm, Eng. Optim., vol. 53, no. 11, pp. 1922–1945, 2021. DOI: 10.1080/0305215X.2020.1839442.

[2] R.J.R. Pereira, F.A. de Almeida, and G.F. Gomes, A multiobjective optimization parameters applied to additive manufacturing: doe-based approach to 3d printing, Structures., vol. 55, pp. 1710–1731, 2023. Elsevier DOI: 10.1016/j.istruc.2023.06.136.

[3] M.P. Bendsøe, and O. Sigmund, Topology Optimization: Theory, Methods, and Applications, Berlin, Heidelberg: Springer-Verlag, 2003.

[4] F. Buonamici, M. Carfagni, R. Furferi, L. Governi, and Y. Volpe, Generative design: an explorative study, CADandA., vol. 18, no. 1, pp. 144–155, 2020. DOI: 10.14733/cadaps.2021.144-155.

[5] I. Gibson, D. Rosen, B. Stucker, and M. Khorasani, Additive Manufacturing Technologies, 3rd ed., Springer, 2021.

[6] N. Volpato, Manufatura Aditiva: Tecnologias e Aplicações da Impressão 3D, São Paulo: blucher, 2017.

[7] L.J. Gibson, and M.F. Ashby, Cellular Solids: Structure and Properties, 2nd ed., Cambridge: Cambridge University Press, 1997.

[8] A. Panesar, M. Abdi, D. Hickman, and I. Ashcroft, Design and manufacturing of lattice structures: a review, Appl Sci., vol. 8, no. 2, pp. 294, 2018.

[9] I. Maskery, N.T. Aboulkhair, A.O. Aremu, C.J. Tuck, and I.A. Ashcroft, Mechanical properties of ti-6al-4v selective laser melted parts with body-centred-cubic lattices of varying strut diameter, Mater. Sci. Eng, A., vol. 714, pp. 211–224, 2018.

[10] L.C. Silva, Fadiga em Ligas Ti-6Al-4V Processadas Por Manufatura Aditiva., Tese (doutorado em engenharia mecânica), Escola Politécnica da Universidade de São Paulo (USP), São Paulo, 2023.

[11] O. Al-Ketan, and R.K. Abu Al-Rub, Multifunctional mechanical metamaterials based on triply periodic minimal surface lattices, Adv. Eng. Mater., vol. 21, no. 10, pp. 1900524, 2019. DOI: 10.1002/adem.201900524.

[12] L. Zhang, S. Feih, S. Daynes, S. Chang, M. Wang, J. Wei, and W. Lu, Fatigue behavior of gyroid cellular structures manufactured by selective laser melting, Mater Des., vol. 198, pp. 109332, 2020.

[13] M. Leary, L. Merli, F. Torti, M. Mazur, and M. Brandt, Optimal topology for additive manufacture: a method for enabling additive manufacture of support-free structures, Mater Design., vol. 63, pp. 678–690, 2014. DOI: 10.1016/j.matdes.2014.06.015.

[14] D. Herzog, V. Seyda, E. Wycisk, and C. Emmelmann, Additive manufacturing of metals, Acta Mater., vol. 117, pp. 371–392, 2016. DOI: 10.1016/j.actamat.2016.07.019.

[15] Y. Tang, G. Dong, Q. Zhou, and Y. Zhao, Design and optimization of hybrid lattice structures for additive manufacturing, Virtual Phys. Prototyp., vol. 14, no. 4, pp. 342–363, 2019.
[16] J. Walczak, and W. Szaniec, Generative design in the context of additive manufacturing, Manage Product Engin Rev., vol. 9, no. 1, pp. 112–124, 2018.
[17] M.K. Thompson, G. Moroni, T. Vaneker, G. Fadel, R.I. Campbell, I. Gibson, A. Bernard, J. Schulz, P. Graf, B. Ahuja, and F. Martina, Design for additive manufacturing: trends, opportunities, considerations, and constraints, CIRP Ann., vol. 65, no. 2, pp. 737–760, 2016. DOI: 10.1016/j.cirp.2016.05.004.
[18] V. Mishra, S. Negi, and S. Kar, Fdm-based additive manufacturing of recycled thermoplastics and associated composites, J. Mater. Cycles Waste Manag., vol. 25, no. 2, pp. 758–784, 2023. DOI: 10.1007/s10163-022-01588-2.
[19] W. Gao, Y. Zhang, D. Ramanujan, K. Ramani, Y. Chen, C.B. Williams, C.C. Wang, Y.C. Shin, S. Zhang, and P.D. Zavattieri, The status, challenges, and future of additive manufacturing in engineering, Comput-Aided. Des., vol. 69, pp. 65–89, 2015. DOI: 10.1016/j.cad.2015.04.001.
[20] X. Yan, W. Rao, and Y. Li, Gyroid structures in additive manufacturing: a review, Mater. Today Commun., vol. 31, pp. 103325, 2022.
[21] M.B. Francisco, J.L.J. Pereira, G.A.V.B. Vasconcelos, S.S. da Cunha, Jr, and G.F. Gomes, Multiobjective design optimization of double arrowhead auxetic model using lichtenberg algorithm based on metamodelling. In: Structures, vol. 45, Amsterdam: Elsevier, pp. 1199–1211, 2022.
[22] J. Cypriano, et al., Caracterização mecânica e microestrutural da liga alsi10mg processada por fusão seletiva a laser, Matéria (Rio De Janeiro)., vol. 25, no. 2, 2020.
[23] A.A. Couto, et al., Fabricação da liga ti-6al-4v por manufatura aditiva: uma análise bibliométrica. In Anais Do X Congresso Brasileiro de Engenharia de Produção, Ponta Grossa, PR, CONBREPRO, 2019.
[24] J. Dalenogare, Estudo da influência dos parâmetros de processo nas propriedades do aço 316l fabricado por slm, dissertação (mestrado em engenharia mecânica), Universidade Federal do Rio Grande do Sul (UFRGS), Porto Alegre, 2020.
[25] MakerVerse, Metal powder costs in 3d printing, 2025. Available on MakerVerse.
[26] AMPOWER, "Additive manufacturing market 2024", 2024. Available on AMPOWER.
[27] M.B. Francisco, J.L.J. Pereira, G.A. Oliver, F.H.S. da Silva, S.S. da Cunha, Jr, and G.F. Gomes, Multiobjective design optimization of cfrp isogrid tubes using sunflower optimization based on metamodel, Comp Struct., vol. 249, pp. 106508, 2021. DOI: 10.1016/j.compstruc.2021.106508.
[28] General Electric, Ge jet engine bracket challenge, 2013. Disponível em GrabCAD.